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Market Maker or Informed Trader: Who Drive the Relationship Between Option Trading and Underlying Returns? Evidence From Shanghai Stock Exchange 50 ETF Options

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ABSTRACT

Using option order imbalance as a proxy for market makers' inventory pressure, we identify a distinct return reversal pattern between option trading activity and underlying asset returns. Specifically, call (put) order imbalances are contemporaneously positively (negatively) associated with underlying returns, followed by rapid reversals in the subsequent period. This reversal stems primarily from temporary price pressures caused by market makers' delta-hedging activities and remains robust after controlling for multiple factors and across various option classifications. Two empirical scenarios reinforce that option order imbalance reflects market makers' inventory risks rather than informed trading. Additional analyses—including event studies, out-of-sample tests, examinations of dynamic hedging behavior, and panel regressions with expanded SSE-listed exchange traded fund option data—further substantiate these findings. Overall, our results emphasize the critical role of market makers as a noninformational trading channel, significantly shaping the relationship between option trading and underlying asset prices.

JEL Classification: G13, G14, D53

1 | Introduction

Options are pivotal financial derivatives, and understanding how trading in options affects the price dynamics of underlying assets remains a central issue in empirical asset pricing research (Bollen and Whaley 2004; Ni et al. 2008). A growing literature emphasizes that options are not redundant securities and highlights two primary channels through which option markets influence underlying asset returns (Cao et al. 2023).

The first is the informed trading channel. Investors possessing private information about future price movements are incentivized to transact in option markets, given their inherent leverage, lower transaction costs, and fewer short-selling

All authors contributed equally to this study.

constraints relative to underlying asset markets (Black 1975; Easley et al. 1998; Ofek et al. 2004; Pan and Poteshman 2006). Because private information permeates market prices, option order flows can serve as informative predictors of subsequent returns, underscoring the role of options in price discovery.

The second channel is inventory-driven delta-hedging. Market makers and liquidity providers, typically positioned opposite customer trades, absorb inventory risk—particularly in less liquid markets (Avellaneda and Lipkin 2003; Ni et al. 2005 ; Golez and Jackwerth 2012). To manage exposure, these market makers dynamically hedge their positions in the underlying asset markets to maintain delta neutrality. Such hedging actions trigger temporary price pressures, leading to short-term return reversals once positions are subsequently unwound.

Focusing on the Shanghai Stock Exchange (SSE) 50 exchange traded fund (ETF) options market, this study provides novel empirical evidence on how option trading influences underlying price dynamics. Our analysis places particular emphasis on the hedging activities of market makers in China's emerging options market. While studies to date have generally concentrated on developed economies—particularly the US market—this study addresses whether similar mechanisms prevail in emerging derivatives markets, which operate under distinct institutional frameworks and investor behaviors.

China's options market, launched in 2015 with the introduction of SSE 50 ETF options, is relatively young and highly regulated compared to its US and European counterparts. Strict rules and investor eligibility criteria significantly shape market structure. Access to trading requires brokerage approval contingent on investor experience, risk tolerance, and capital adequacy. Regulators impose stringent caps on trading volumes and position sizes, especially for new entrants, with incremental increases permitted only as investors accumulate experience.[1] Additional restrictions, such as circuit breakers and maximum order sizes, further constrain investor trading activities. Collectively, these regulatory policies substantially curtail opportunities to exploit private information.

China also employs a distinctive competitive market maker system to provide liquidity. Unlike the US market, which includes designated market makers as well as various liquidity providers, the Chinese market incorporates primary and general market makers subject to distinct quoting obligations and policy incentives. Consequently, the Chinese options landscape regularly experiences persistent one-sided order flows and pronounced buy-sell imbalances—features seldom observed in developed markets. Moreover, whereas market makers in developed markets often maintain net short positions (Hu 2014; Fournier and Jacobs 2020), Chinese market makers usually carry net long positions and act primarily as liquidity providers rather than informed traders. This unique environment provides a valuable setting for examining how delta-hedging by market makers influences underlying asset returns.

Following methods used by Chordia and Subrahmanyam (2004) and Christoffersen et al. (2018), we construct option order imbalance measures from call and put trades to capture net buying pressure and proxy for the inventory risk borne by market makers. Our empirical analyses investigate both contemporaneous and lagged relationships between these order imbalances and SSE 50 ETF returns. We uncover significant contemporaneous associations—positive for call imbalances and negative for put imbalances—followed by clear reversals in subsequent periods, consistent with the dynamics induced by delta-hedging. These reversal effects remain robust even after controlling for standard risk factors and option-implied variables, and persist across various option categories classified by moneyness, maturity, and trade size.

To further assess the delta-hedging channel, we build on the framework developed by Bollen and Whaley (2004) and provide additional supportive evidence distinguishing inventory-driven explanations from informed trading hypotheses. Meanwhile, we find that return reversal patterns are more pronounced when liquidity conditions deteriorate, as measured by the relative bid-ask spread and absolute illiquidity (ILLIQ), consistent with increased hedging costs. We also examine the impact of significant market events, conduct robustness tests spanning different subperiods, and demonstrate economic relevance through trading strategies in out-of-sample analyses. Furthermore, following Baltussen et al. (2021), we explore market makers' net gamma exposure and document evidence consistent with volatility reductions arising from their hedging activities. Panel-data regression analyses employing additional SSE-listed ETF options further confirm the robustness of our core findings.

Our work is closely related to Zheng and Luo (2024), who investigate dynamic market-maker hedging behavior and intraday reversals in the Chinese commodity options and futures markets. To our knowledge, theirs is the first empirical study of market-maker hedging in China's derivatives markets. However, our study differs in several important respects. First, we focus exclusively on equity options and underlying ETF returns, analyzing daily return reversals aligned with China's T+1 equity settlement rules rather than intraday dynamics. Second, rather than directly observing hedging trades, we infer market-maker behavior from option order imbalances and underlying returns, and additionally examine implications for volatility. Third, while Zheng and Luo (2024) report that enhanced liquidity amplifies intraday return reversals, we find that greater options liquidity mitigates daily return reversal patterns—a seeming contradiction that is reconciled by differences in measurement frequency.

The remainder of the paper is organized as follows. Section 2 reviews the related literature. Section 3 provides an overview of China's options market, data, and variable construction. Section 4 presents our main empirical results and underlying mechanisms. Section 5 offers supplementary analyses and external robustness checks. Section 6 concludes.

2 | Literature Review

Our study relates to several strands of the literature.

First, we contribute to the expanding body of research examining the impact of options trading on underlying asset returns through non-informational channels. For example, Chen et al. (2019) find that order imbalance in deep out-of-the-money index put options can negatively predict market returns, reflecting the constraints faced by financial intermediaries. Similarly, Goncalves-Pinto et al.

(2020) argue that the predictability of option prices for stock returns is primarily driven by price pressure and liquidity effects in the stock market, rather than improved price discovery in the options market. Chordia et al. (2021) further demonstrate that the predictive power of index put option order imbalance can be attributed to retail investors' risk-protection strategies, particularly during periods of heightened market uncertainty.

Second, our findings contribute to the literature on the role of option market makers' inventory risk. Muravyev (2016) demonstrates that inventory risk is a key determinant in option price formation, as market makers demand compensation for deviations from optimal inventory levels. Fournier and Jacobs (2020) show that inventory risk significantly affects option prices by influencing the variance risk premium. The hedging activities of market makers can also have notable cross-market effects; for instance, Sensoy and Omole (2022) find that weekly call order imbalances, through delta-hedging activities, can impact underlying asset prices. Recent research has also begun to consider the aggregate gamma exposure of delta-neutral market makers. Baltussen et al. (2021) construct daily measures of net gamma exposure and show that intraday price momentum can be explained by hedging demands implied by net gamma positions. Using transaction data, Ni et al. (2021) find a negative relationship between market makers' net gamma imbalance and the volatility of the underlying assets.

Third, our research is closely related to studies focusing on the Chinese options market. Prior work has highlighted the predictive power of options trading activity for subsequent movements in the underlying assets, often emphasizing the informational transmission channel. For instance, Luo et al. (2020) find that option order imbalances can predict returns of underlying A-share stocks, attributing this to informed trading in the A-H share market. Luo et al. (2022) show that both positive (buying calls and selling puts) and negative (selling calls and buying puts) trading volumes in SSE 50 ETF options significantly impact SSE 50 Index futures prices. Liu et al. (2024) further confirm the forecasting ability of option-implied ambiguity for future SSE 50 ETF returns. Additionally, several studies have examined the influence of options on the underlying's volatility (Arkorful et al. 2020; Sui et al. 2021; Wu, Liu and Feng 2022; Wu, Liu, Yuan et al. 2022; Gong et al. 2024).

3 | Market and Data

3.1 | An Overview of the Chinese A-Share ETF Options Market

Approved by the China Securities Regulatory Commission (CSRC), SSE 50 ETF options were officially listed on February 9, 2015, marking the launch of China's first ETF options product in the A-share market. This remained the sole stock index ETF option available until December 23, 2019, when the CSI 300 ETF option was introduced. By the end of 2024, the SSE had listed four ETF option products: SSE 50 ETF Options, CSI 300 ETF Options, CSI 500 ETF Options, and STAR 50 ETF Options.

China's stock index ETF options market has attracted significant investor attention, with trading volumes rising rapidly after its launch and making it one of the most active options markets in the world. According to the 2024 Annual Report on the Stock Option Market published by the SSE, in 2024, the average daily trading volume reached 4.89 million contracts, with an average daily open interest of 6.14 million contracts. Specifically, for the SSE 50 ETF options, the average daily trading volume was 1.39 million contracts and the average daily open interest was 1.79 million contracts, corresponding to a turnover rate of approximately 77%. The number of stock option investors also continued to grow steadily, reaching 683,700 accounts by year-end 2024.

China's options market has implemented a market maker system to provide liquidity, adopting a model of "primary market makers as the backbone and general market makers as supplements."[2] For the SSE 50 ETF options, there were 22 market makers at the end of 2024, including 19 primary and 3 general market makers. As in other major global markets, market makers play a dominant role in trading. In 2024, primary market makers recorded an average daily trading volume of 4.43 million contracts (two-way), accounting for 45.31% of the total market trading volume, and an average daily open interest of 2.17 million contracts (two-way), representing 17.98% of the market. Figures 1 and 2 plot the time series of the total number of SSE option accounts, the daily average trading volume, and the share of primary market makers from 2015 to 2024.

3.2 | Data and Variables

This subsection describes our data sources, variable construction, and summary statistics. Our primary data are obtained from the Chinese Stock Market and Accounting Research (CSMAR) and WIND databases, covering the period from February 9, 2015, to February 9, 2025, for a total of 2427 trading days.

We utilize comprehensive option transaction-level data, which contains variables such as option identifier, option type (call or put), trade direction (buyer-initiated or seller-initiated), transaction price and volume, and indicators for whether a trade is opening or closing a position. Each option's strike price and expiration date are identified via its option ID, allowing us to compute moneyness and days to expiration.

While our main analysis focuses on SSE 50 ETF options, we also include CSI 300 ETF and CSI 500 ETF options in robustness checks to ensure the generality of our findings.[3]

3.2.1 | Sample Screening Criteria

We apply the following filters to construct the final sample:

1. Exclude options that are not traded during continuous trading sessions (09:30–11:30 and 13:00–14:57).
2. Exclude options with zero trading volume, or with less than 7 days or more than 100 days to maturity.
3. Exclude options with absolute delta less than 0.02 or greater than 0.98.

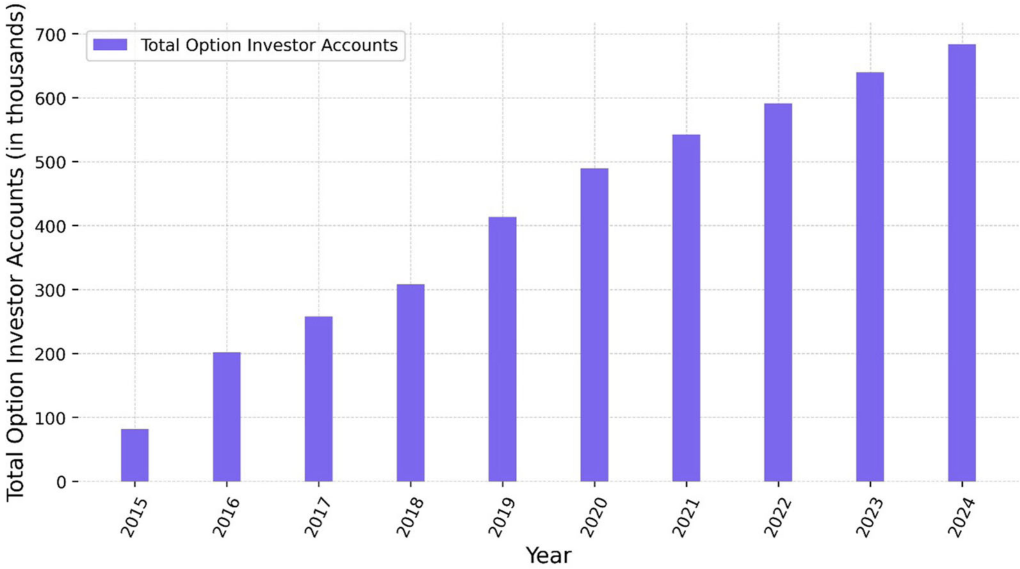

FIGURE 1 | Total option investor accounts.

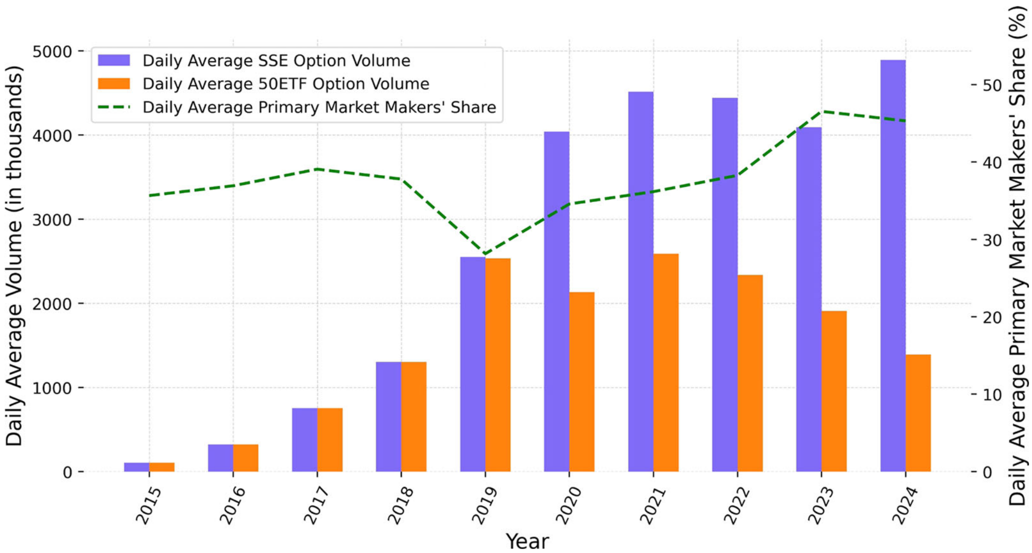

FIGURE 2 | The daily average option trading volume and primary market maker's share.

4. Exclude options that do not meet standard no-arbitrage constraints.[4]

3.2.2 | Order Imbalance Measure

We define the daily call and put order imbalance as the summation of position-opening buyer-initiated trading volume less seller-initiated trading volume, divided by the total daily trading volume of position-opening trades, given as the following equations:

$$\mathrm{CallOIB_t} = \frac{\sum_i \left(\mathrm{BuyVol}_{i,t}^{\mathrm{Call}} - \mathrm{SellVol}_{i,t}^{\mathrm{Call}}\right)}{\sum_i \left(\mathrm{BuyVol}_{i,t}^{\mathrm{Call}} + \mathrm{SellVol}_{i,t}^{\mathrm{Call}}\right)}, \mathrm{PutOIB}_t = \frac{\sum_i \left(\mathrm{BuyVol}_{i,t}^{\mathrm{Put}} - \mathrm{SellVol}_{i,t}^{\mathrm{Put}}\right)}{\sum_i \left(\mathrm{BuyVol}_{i,t}^{\mathrm{Put}} + \mathrm{SellVol}_{i,t}^{\mathrm{Put}}\right)}. \tag{1}$$

For the call option contract i on day t, $\mathrm{BuyVol}_{i,t}^{\mathrm{Call}}$ represents the trading volume initiated by the buyer, and $\mathrm{SellVol}_{i,t}^{\mathrm{Call}}$ represents the trading volume initiated by the seller. The numerator of the call order imbalance, $\mathrm{CallOIB}_t$, measures the hedging demand for the underlying asset of market makers, standardized by the total daily trading volume. PutOIB_t is the put order imbalance constructed similarly to $\mathrm{CallOIB}_t$. For robustness, we also construct delta-weighted call and put order imbalance as proxies for inventory pressure on market makers following Christoffersen et al. (2018). Furthermore, the findings of this paper remain consistent even when using K/S (strike-to-spot price ratio), another commonly employed metric for aggregating different option contracts, as an alternative weighting scheme.[5]

Following Chordia and Subrahmanyam (2004), we define the daily return of the underlying SSE 50 ETF as the open-to-close return, calculated as the log difference between the daily closing and opening price $\mathrm{Return}_t = \mathrm{In}\left(P_t^{\mathrm{Close}}\right) - \mathrm{In}\left(P_t^{\mathrm{Open}}\right)$, where

P_t^{Open} and P_t^{Close} denote the opening and closing price on day t, respectively.[6]

In regression, we include several control variables. We first construct two macroeconomic factors, TermSpread and MarketHV. TermSpread is calculated as the difference between the 10-year and 6-month maturity Chinese Treasury yields.[7] MarketHV denotes the historical volatility of the SSE Composite Index, computed as the rolling standard deviation of daily returns over the past 30 days. We then incorporate two micro-economic factors, Turnover and ILLIQ. Turnover refers to the turnover rate of underlying SSE 50 ETF which reflects investor sentiment. And ILLIQ is the illiquidity measure of underlying asset, as proposed by Amihud (2002).

Following the literature, we also construct a set of option trading implied variables, which have been documented to possess predictive ability for underlying asset returns, composed of PCR, OSR, IVS, SKEW, POV and NOV. PCR is the put-to-call ratio, defined as the position-opening buyer-initiated put trading volume divided by the total position-opening buyer-initiated trading volume, following Pan and Poteshman (2006). OSR is the option-to-stock volume ratio proposed by Johnson and So (2012), defined as the logarithm of the ratio of option trading volume to the trading volume of the underlying ETF, where the option volume is adjusted by a contract multiplier of 10,000. IVS is the implied volatility spread, measured as the volume-weighted average difference between strike- and maturity-matched call-implied and put-implied volatilities, following Cremers and Weinbaum (2010). SKEW is the implied volatility smirk proposed by Xing et al. (2010), as the difference between the implied volatilities of OTM puts and ATM calls. POV and NOV denote the positive and negative option volumes, respectively, measured by the number of transactions, as proposed by Luo et al. (2022). Specifically, POV is the total number of transactions involving call purchases and put writings, whereas NOV comprises put purchases and call writings. Both POV and NOV are standardized to have a mean of zero and a standard deviation of one to facilitate coefficient comparison in regression analysis.

To investigate the effect of market makers' dynamic hedging on underlying volatility, we use 5-min price data of underlying assets to calculate realized volatility, $\mathrm{RV}_t = \sqrt{\sum_{\mathrm{j}} \mathrm{Return}_{t,j}^2}$, where $\mathrm{Return}_{t,j}$ denotes the jth intraday return on day t. We measure the net gamma exposure of market makers based on all the open interest in options, following Baltussen et al. (2021). For a call option contract i on day t, the net gamma exposure is computed as

$$\mathrm{NGE}_{i,t}^{\mathrm{Call}} = \Gamma_{i,t}^{\mathrm{Call}} * \mathrm{OI}_{i,t}^{\mathrm{Call}} * 10,000 * P_t, \quad (2)$$

where $\Gamma_{i,t}^{\mathrm{Call}}$ and $\mathrm{OI}_{i,t}^{\mathrm{Call}}$ are the option's gamma and open interest, respectively, and 10,000 is the adjustment from option contracts to shares, P_t is the price of underlying. For a put option contract j, the net gamma exposure is computed analogously as

$$\mathrm{NGE}_{j,t}^{\mathrm{Put}} = \Gamma_{j,t}^{\mathrm{Put}} * \mathrm{OI}_{j,t}^{\mathrm{Put}} * (-10,000) * P_t. \quad (3)$$

The aggregated net gamma exposure is summed across all option contracts and divided by the market value of the underlying (MV_t):

$$\mathrm{NGE}_t = \frac{\sum_i \mathrm{NGE}_{i,t}^{\mathrm{Call}} + \sum_j \mathrm{NGE}_{j,t}^{\mathrm{Put}}}{\mathrm{MV}_t}. \quad (4)$$

Panel A of Table 1 reports the daily average trading volumes and order imbalances for SSE 50 ETF option contracts. For call options, the buyer-initiated trading volume averages 120,000 contracts per day, while seller-initiated volume averages 163,000 contracts, resulting in an average daily call order imbalance of −13.35%. This negative order imbalance suggests a preference among end users for selling rather than buying calls. Given that market makers act as liquidity providers and take the opposite side of customer trades, this pattern implies that market makers may face positive inventory pressure through net long call exposures. A similar phenomenon is observed for put options, the average put order imbalance is −7.09%. Comparatively, call trading volume exceeds put volume, and the magnitude of order imbalance in calls is notably larger than that for puts. These findings indicate that market makers are subject to greater inventory pressure from call options than from puts.

Disaggregating by contract characteristics, the remaining part of Panel A shows that negative order imbalance is pervasive regardless of moneyness, maturity, or trade size.[8] Out-of-the-money (OTM) and at-the-money (ATM) contracts see higher trading volumes than in-the-money (ITM) options. Similarly, contracts with shorter maturities and larger trade sizes account for the majority of total volume in their respective groups. Notably, unlike Table 1 in Hu (2014)––where buyer-initiated volumes exceed seller-initiated volumes across all categories––our results exhibit the opposite pattern. This distinction may reflect the early-stage nature of China's options market, which is characterized by a relatively high proportion of noise trading and speculative activity. Accordingly, the information content of option order imbalance in China may be lower compared to the more mature market context of Hu (2014).

Panel B of Table 1 presents summary statistics for the main variables used in our analysis. The average daily return and realized volatility of the SSE 50 ETF are 0.045% and 0.996%, respectively. Notably, the skewness and kurtosis of the call order imbalance are both higher than those of the put order imbalance, supporting prior findings that market makers accumulate larger long positions in call options. Furthermore, the mean of the net gamma exposure is positive, consistent with the notion that market makers are net buyers of options. Pearson correlation coefficients (reported in Appendix Table A1) show a strong positive correlation of 0.49 between call order imbalance and underlying returns, and a strong negative correlation of −0.48 between put order imbalane and returns. These coefficients are substantially larger than those for other variables, underscoring the central relationship between option order imbalance and underlying returns.

TABLE 1 | Descriptive statistics and autocorrelation of order imbalance.

Panel A: Average daily option trading volume and order imbalance

	Total	OTM	ATM	ITM	Short	Middle	Long	Small	Medium	Large
Call Buy	120	56	42	22	84	31	5	4	15	101
Call Sell	163	72	59	32	113	42	7	5	22	136
CallOIB	−13.347	−9.652	−15.016	−17.886	−12.640	−14.554	−16.033	−15.150	−17.744	−12.394
Put Buy	104	51	37	17	71	28	5	3	15	86
Put Sell	127	61	45	21	89	32	6	4	18	105
PutOIB	−7.090	−5.421	−8.100	−7.713	−8.138	−4.864	−5.900	−9.671	−9.774	−6.188

Panel B: Summary statistics of variables

	N	Mean	Min	Q10	Median	Q90	Max	Std	Skewness	Kurtosis
Return	2192	0.045	−8.711	−1.193	0.033	1.402	9.893	1.250	0.037	7.102
CallOIB	2192	−13.325	−50.515	−24.880	−14.637	0.292	60.872	10.465	1.169	3.703
PutOIB	2192	−7.272	−41.407	−20.194	−8.331	6.989	42.527	10.926	0.652	0.969
TermSpread	2191	−0.000	−0.242	−0.041	0.000	0.039	0.310	0.038	0.040	5.931
MarketHV	2162	17.919	4.491	9.431	14.471	30.007	63.495	10.313	2.035	4.651
Turnover	2192	4.140	0.636	1.612	3.207	6.706	69.652	4.079	5.764	54.485
ILLIQ	2162	25.856	8.467	16.681	24.075	35.973	88.109	10.222	2.571	10.527
PCR	2192	46.876	24.872	39.795	46.967	53.147	72.534	5.531	0.029	1.387
OSR	2192	265.000	−334.273	135.089	295.416	379.281	469.805	128.010	−2.040	4.599
IVS	2192	−0.965	−97.797	−6.323	0.431	3.421	36.171	6.724	−5.153	51.124
SKEW	2192	−2.316	−21.378	−5.470	−2.012	0.667	23.452	3.046	−0.505	8.566
POV	2192	1.907	−165.420	−129.236	0.158	128.725	526.254	102.582	0.643	0.696
NOV	2192	1.795	−167.312	−130.007	1.612	124.552	485.020	102.615	0.597	0.582
RV	2192	0.996	0.279	0.518	0.859	1.555	7.055	0.612	4.066	26.955
NGE	2192	27.862	−13.719	4.598	22.169	58.972	155.283	23.023	1.388	2.428

Panel C: Autocorrelation coefficients of option order imbalance

	Lag 1	Lag 2	Lag 3	Lag 4	Lag 5	Lag 6	Lag 7	Lag 8	Lag 9	Lag 10
CallOIB	0.224***	0.154***	0.116***	0.066**	0.025	0.023	0.069**	0.035	0.046*	0.052**
	(0.028)	(0.028)	(0.027)	(0.027)	(0.027)	(0.029)	(0.034)	(0.027)	(0.027)	(0.025)
PutOIB	0.290***	0.119***	0.112***	0.046*	−0.009	0.027	0.032	0.046*	0.037	0.041*
	(0.024)	(0.024)	(0.026)	(0.027)	(0.026)	(0.024)	(0.026)	(0.026)	(0.025)	(0.024)

Note: Panel A of the table reports average daily position-opening options trading volumes (expressed in thousands of contracts) and order imbalance (expressed in percentage terms). Total indicates the average trading volume of all signed trades. Call Buy and Call Sell represent the call options trading volumes of buyer-initiated and seller-initiated, respectively. And Put Buy and Put Sell follow analogous definitions for put options. OTM (out-of-the-money), ATM (at-the-money), and ITM (in-the-money) classify options by moneyness. Short, Middle and Long classify options by time to maturity and Small, Medium and Large classify options by trade size. Panel B reports summary statistics of the variables, and all variables are expressed in percentage terms. Return denotes the daily open-to-close return of underlying. CallOIB and PutOIB denote the call and put order imbalance, respectively. TermSpread is the first difference of the spread between the 10-year and 6-month maturity Chinese Treasury yields. MarketHV denotes the historical volatility of the Shanghai Stock Exchange Composite Index. Turnover and ILLIQ denote the daily turnover rate and the illiquidity of the underlying SSE 50 ETF, respectively. PCR, OSR, IVS, and SKEW represent the put-to-call ratio, option-to-ETF volume ratio, implied volatility spread and implied volatility smirk, respectively. POV and NOV denote the daily positive and negative option volume, respectively, standardized (using z-score normalized) to ensure comparability. RV and NGE represent the realized volatility of underlying ETF and net gamma exposure of option market makers, respectively. Panel C reports the autocorrelation coefficients of call and put order imbalance up to 10 lags. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

Panel C of Table 1 presents the autocorrelation coefficients for both call and put order imbalances. The results reveal significant positive autocorrelation in both series, consistent with the findings of Chordia and Subrahmanyam (2004), who argue that persistent inventory pressure arises from autocorrelated order flows.

4 | Empirical Results

This section investigates the relationship between underlying returns and option order imbalances by estimating both contemporaneous and lagged effects. To ensure the robustness of our findings, we control for a variety of factors and also conduct

TABLE 2 | Open-to-close return and option order imbalance.

	Contemporaneous effects			Lagged effects			Reversal effects		
	(1)	**(2)**	**(3)**	**(4)**	**(5)**	**(6)**	**(7)**	**(8)**	**(9)**
CallOIB_t	0.060***		0.051***				0.077***		0.056***
	(0.004)		(0.004)				(0.004)		(0.004)
CallOIB_{t-1}				0.005*		0.005*	−0.039***		−0.016***
				(0.003)		(0.003)	(0.004)		(0.004)
PutOIB_t		−0.055***	−0.046***					−0.070***	−0.046***
		(0.003)	(0.003)					(0.003)	(0.004)
PutOIB_{t-1}					−0.004	−0.004		0.035***	0.010**
					(0.004)	(0.004)		(0.004)	(0.004)
Return_{t-1}	−0.003	−0.001	−0.008	−0.014	−0.010	−0.034	0.156***	0.141***	0.098**
	(0.031)	(0.029)	(0.029)	(0.036)	(0.040)	(0.047)	(0.035)	(0.036)	(0.044)
Intercept	0.846***	−0.343***	0.400***	0.114**	0.017	0.088	0.544***	−0.209***	0.314***
	(0.067)	(0.041)	(0.071)	(0.047)	(0.043)	(0.054)	(0.057)	(0.042)	(0.063)
Adj. R^2	0.244	0.228	0.397	0.001	0.000	0.001	0.303	0.281	0.413
Obs.	2426	2426	2426	2426	2426	2426	2426	2426	2426

Note: The table reports the results of the following regression:

$$\begin{aligned}\text{Return}_t = \alpha + \beta_0\text{CallOIB}_t + \beta_1\text{CallOIB}_{t-1} + \gamma_0\text{PutOIB}_t + \gamma_1\text{PutOIB}_{t-1}\\ + \phi\text{Return}_{t-1} + \varepsilon_t,\end{aligned}$$

where Return_t is the open-to-close return of the SSE 50 ETF on day t. CallOIB_t and CallOIB_{t-1} represent the call order imbalance on day t and $t-1$, respectively. And PutOIB_t and PutOIB_{t-1} follow analogous definitions for put options. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

a heterogeneity analysis by classifying options into different categories.

4.1 | Baseline Results of Return Reversal

Following Chordia and Subrahmanyam (2004), we start by estimating the following regression to analyze the contemporaneous impact of market makers' inventory risk on underlying asset return.

$$\text{Return}_t = \alpha + \beta_0\text{CallOIB}_t + \gamma_0\text{PutOIB}_t + \phi\text{Return}_{t-1} + \varepsilon_t, \tag{5}$$

where Return_t represents the SSE 50 ETF return. CallOIB_t and PutOIB_t represent call and put order imbalance, respectively. We then estimate the effects of lagged order imbalance by the following regression,

$$\text{Return}_t = \alpha + \beta_1\text{CallOIB}_{t-1} + \gamma_1\text{PutOIB}_{t-1} + \phi\text{Return}_{t-1} + \varepsilon_t. \tag{6}$$

Finally, we combine the contemporaneous and lagged terms together and estimate the following full model:

$$\begin{aligned}\text{Return}_t = \alpha + \beta_0\text{CallOIB}_t + \beta_1\text{CallOIB}_{t-1}\\ + \gamma_0\text{PutOIB}_t + \gamma_1\text{PutOIB}_{t-1}\\ + \phi\text{Return}_{t-1} + \varepsilon_t.\end{aligned}, \tag{7}$$

Table 2 presents the estimation results. Columns 1–3 show a significant positive contemporaneous relationship between call order imbalance and underlying returns, and a negative relationship for put order imbalance. Columns 4–6 report that lagged call order imbalance remains positively associated with subsequent returns. Columns 7–9 indicate that call order imbalance has a positive contemporaneous effect, followed by a rapid reversal in the subsequent period. Specifically, in Column 9, a one standard deviation increase in call order imbalance leads to a 0.59% (0.056×10.465) rise in contemporaneous returns and a 0.18% (-0.016×10.465) decline in the following period. Put order imbalance shows a symmetric pattern, with a negative contemporaneous effect that reverses in the next period, though its magnitude is notably smaller.

For interpretative clarity, in the model that jointly estimates the effects of contemporaneous and lagged option order imbalances, we define the coefficient on contemporaneous order imbalance as the "contemporaneous effect" and the lagged coefficient as the "reversal effect." For example, in Column 9, the return reversal effect for call order imbalance is −0.016, which is greater in absolute value than that for put order imbalance (0.010).

We interpret these results in the context of market makers' hedging activities, as outlined by Chordia and Subrahmanyam (2004). The significant contemporaneous effects in Columns 1–3 can be attributed to delta-hedging: when call order imbalance rises—indicating increased buyer-initiated call trades by end users—market makers, as counterparties, accumulate short call positions. They hedge this exposure by purchasing the underlying asset, exerting upward pressure on prices. A similar but opposite effect applies for puts. The contemporaneous effect of call order imbalance is stronger than that of puts, potentially due to higher trading volumes and larger imbalances in calls, resulting in greater inventory risk for market makers. When both call and put order imbalances are included in the regression (Column 3), the magnitudes of their effects attenuate because their hedging impacts partially offset each other.

The lagged effects, as shown in Columns 4–6, are statistically significant only for call order imbalance, consistent with calls generating greater inventory pressure. These lagged effects are weaker than their contemporaneous counterparts, possibly because order imbalance autocorrelation is less pronounced for options than equities, or because market makers in the SSE 50 ETF options market face stricter inventory management and resolve their positions more quickly—often within the same trading day.

In the full model (Columns 7–9), the lagged coefficients reverse sign relative to the contemporaneous ones, supporting the price pressure hypothesis. This reversal is due to the autocorrelation of order imbalance and the transitory impact of market makers' hedging trades on prices; after the temporary distortion, prices revert to their fundamental values. Consistent with earlier results, call and put order imbalances reflect opposing hedging pressures, resulting in smaller return reversal effects in the full model than in models estimated separately for calls or puts. For instance, in Column 7, the return reversal effect for call order imbalance is −0.039, greater in magnitude than the -0.016 estimated in Column 9.

In summary, we document a clear return reversal effect driven by market makers' delta-hedging activities: the impact of option order imbalance on underlying returns reverses in the subsequent period. This result remains robust under alternative specification, including daily close-to-close returns and weighted order imbalance measures, as shown in Table A2 of the Appendix.

4.2 | Robust Check by Controlling Other Risk Factors

A potential concern regarding the documented return reversal effect is that option order imbalance might merely proxy for exposures to other risk factors. To allay this concern, we first incorporate two macroeconomic indicators, TermSpread and MarketHV. Additionally, we control for investors' sentiment and underlying illiquidity using Turnover and ILLIQ, respectively. The results of Columns 1–3 in Table 3 demonstrate that the return reversal effect remains statistically significant after controlling for both macroeconomic and microeconomic factors. Moreover, as discussed in Hu (2014), option flows arising from informed trading also affect market makers' hedge demand. Therefore, we control for six option trading-implied variables, PCR, OSR, IVS, SKEW, POV and NOV. The results reported in Columns 4–6 reveal that the return reversal effect retains even when incorporating these controls, suggesting that option order imbalance contains incremental information about the underlying market beyond these option implied variables. Columns 7–9 show that our conclusion remains consistent after simultaneously controlling for all risk factors.

4.3 | Heterogeneity Analyses

To examine whether our results depend on the features of options, we conduct a series of heterogeneity analyses by stratifying option trades based on moneyness, time to maturity, and trade size. First, it is widely acknowledged that informed traders prefer options with high leverage. However, the results in Panel A of Table 4 indicate that the return reversal effect exists across different leverage levels, and the magnitude is strongest for the most traded OTM and ATM options. A plausible explanation is that the most traded OTM and ATM options are more likely to influence market makers' inventory management. Second, regarding time to maturity, we continue to find that the reversal effect persists across all maturity categories, as shown in Panel B. Moreover, the strength of the return reversal increases progressively as time to maturity shortens, which is consistent with prior literature suggesting that option market makers tend to engage in more hedging activities as expiration approaches. Third, with respect to trade size, the results in Panel C demonstrate that the reversal effect remain consistent across all trade size classifications as well. The magnitude of the return reversal effect is more pronounced in medium and large trades, which may be attributed to the fact that larger trade volumes exert a more substantial impact on market makers' inventory positions compared to smaller trades.

4.4 | Mechanisms Behind the Return Reversal

The preceding analysis documents a significant return reversal associated with option order imbalance, providing empirical support for the hypothesis that market makers' hedging demand, driven by inventory risk, influences underlying asset returns. In this subsection, we further seek to distinguish whether the observed relationship between the underlying and options is primarily attributable to hedging-induced price pressure as opposed to informed trading activities. To this end, we design two empirical scenarios aimed at disentangling these mechanisms.

4.4.1 | The Role of Order Imbalance in Shaping the Implied Volatility Function (IVF)

Bollen and Whaley (2004) propose a framework to examine how order imbalance shapes the IVF. According to their argument, order flow can influence the IVF through two potential channels: informed trading and inventory pressure, each operating via distinct mechanisms.

Under the learning hypothesis, order flow reflects informed trading by investors—such as the arrival of private information or changing expectations regarding future volatility. In such cases, market makers interpret order flow as informative and accordingly adjust option prices, resulting in changes in implied volatility. Importantly, because these expectation shifts impart a permanent effect on prices, changes in implied volatility should display little to no autocorrelation; that is, current changes should not systematically predict future changes.

TABLE 3 | Open-to-close return and option order imbalance after controlling various factors.

	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)
CallOIB_t	0.083***		0.058***	0.081***		0.054***	0.085***		0.057***
	(0.004)		(0.004)	(0.004)		(0.004)	(0.004)		(0.004)
CallOIB_{t-1}	−0.040***		−0.016***	−0.041***		−0.020***	−0.042***		−0.018***
	(0.004)		(0.004)	(0.004)		(0.004)	(0.004)		(0.004)
PutOIB_t		−0.075***	−0.050***		−0.076***	−0.049***		−0.078***	−0.051***
		(0.004)	(0.004)		(0.003)	(0.004)		(0.003)	(0.004)
PutOIB_{t-1}		0.032***	0.009**		0.032***	0.011***		0.033***	0.011**
		(0.004)	(0.005)		(0.004)	(0.004)		(0.004)	(0.005)
TermSpread_{t-1}	−0.704	−0.677	−0.672				−0.588	−0.526	−0.477
	(0.908)	(0.907)	(0.879)				(0.780)	(0.758)	(0.734)
MarketHV_{t-1}	−0.006	0.004	−0.000				−0.004	−0.002	−0.002
	(0.005)	(0.004)	(0.005)				(0.005)	(0.005)	(0.005)
Turnover_{t-1}	0.003	0.021	0.016				0.030*	0.025	0.029
	(0.014)	(0.014)	(0.013)				(0.018)	(0.020)	(0.018)
ILLIQ_{t-1}	−0.003	0.012***	0.002				−0.002	0.004	0.000
	(0.004)	(0.004)	(0.004)				(0.004)	(0.005)	(0.004)
PCR_{t-1}				−0.012*	−0.022***	−0.011*	−0.012*	−0.021***	−0.009
				(0.006)	(0.006)	(0.006)	(0.006)	(0.006)	(0.006)
OSR_{t-1}				0.001**	−0.001*	−0.000	0.002***	0.000	0.001
				(0.000)	(0.000)	(0.000)	(0.001)	(0.001)	(0.001)
IVS_{t-1}				−0.009	−0.025**	−0.016	−0.013	−0.025**	−0.017
				(0.012)	(0.011)	(0.011)	(0.012)	(0.012)	(0.011)
SKEW_{t-1}				−0.008	−0.001	-0.014	-0.002	0.001	−0.008
				(0.017)	(0.018)	(0.016)	(0.014)	(0.014)	(0.013)
POV_{t-1}				0.006*	0.006*	0.003	0.006*	0.005	0.001
				(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
NOV_{t-1}				−0.005	−0.007*	−0.003	−0.006*	−0.006*	−0.002
				(0.003)	(0.003)	(0.003)	(0.003)	(0.003)	(0.003)
Return_{t-1}	0.161***	0.135***	0.102**	0.123**	0.062	0.098*	0.118**	0.071	0.097*
	(0.040)	(0.039)	(0.048)	(0.056)	(0.052)	(0.056)	(0.053)	(0.050)	(0.054)
Intercept	0.811***	-0.742***	0.184	0.910***	0.881***	0.718**	0.716**	0.456	0.400
	(0.135)	(0.117)	(0.122)	(0.288)	(0.259)	(0.279)	(0.333)	(0.347)	(0.335)
Adj. R^2	0.308	0.304	0.426	0.323	0.328	0.426	0.326	0.329	0.430
Obs.	2161	2161	2161	2191	2191	2191	2161	2161	2161

Note: This table reports the results of the following regression:

$$\text{Return}_t = \alpha + \beta_0\text{CallOIB}_t + \beta_1\text{CallOIB}_{t-1} + \gamma_0\text{PutOIB}_t + \gamma_1\text{PutOIB}_{t-1} + \lambda\text{Controls}_{t-1} + \phi\text{Return}_{t-1} + \varepsilon_t,$$

where Return_t is the open-to-close return of the SSE 50 ETF on day t. CallOIB_t and CallOIB_{t-1} represent the call order imbalance on day t and $t-1$, respectively. And PutOIB_t and PutOIB_{t-1} follow analogous definitions for put options. The control variables, $\lambda\text{Controls}_{t-1}$, are defined as follows:

$$\lambda\text{Controls}_{t-1} = \lambda_1\text{TermSpread}_{t-1} + \lambda_2\text{MarketHV}_{t-1} + \lambda_3\text{Turnover}_{t-1} + \lambda_4\text{ILLIQ}_{t-1} + \lambda_5\text{PCR}_{t-1} + \lambda_6 OSR_{t-1} + \lambda_7 IVS_{t-1} + \lambda_8\text{SKEW}_{t-1} + \lambda_9\text{POV}_{t-1} + \lambda_{10}\text{NOV}_{t-1}$$

TermSpread is the first difference of the spread between the 10-year and 6-month maturity Chinese Treasury yields. MarketHV denotes the historical volatility of the Shanghai Stock Exchange Composite Index. Turnover and ILLIQ denote the daily turnover rate and the illiquidity of the underlying SSE 50 ETF, respectively. PCR, OSR, IVS and SKEW represent the put-to-call ratio, option-to-ETF volume ratio, implied volatility spread and option volatility smirk, respectively. POV and NOV denote the daily positive and negative option volume respectively. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

TABLE 4 | Open-to-close return and option order imbalance among different categories.

Panel A: Moneyness									
		OTM			**ATM**			**ITM**	
$CallOIB_t$	0.067***		0.050***	0.062***		0.048***	0.042***		0.033***
	(0.003)		(0.003)	(0.004)		(0.004)	(0.003)		(0.003)
$CallOIB_{t-1}$	−0.032***		−0.018***	−0.026***		−0.016***	−0.017***		−0.011***
	(0.003)		(0.003)	(0.003)		(0.003)	(0.003)		(0.003)
$PutOIB_t$		−0.062***	−0.042***		−0.050***	−0.035***		−0.029***	−0.022***
		(0.003)	(0.003)		(0.003)	(0.003)		(0.002)	(0.002)
$PutOIB_{t-1}$		0.026***	0.010***		0.018***	0.008***		0.006***	0.002
		(0.003)	(0.003)		(0.003)	(0.003)		(0.002)	(0.002)
$Return_{t-1}$	0.112**	0.043	0.086*	0.097*	0.027	0.085	0.046	0.046	0.073
	(0.048)	(0.045)	(0.046)	(0.056)	(0.054)	(0.058)	(0.051)	(0.053)	(0.053)
Intercept	0.142	0.253	−0.088	0.556	0.254	0.509	0.467	0.107	0.455
	(0.347)	(0.340)	(0.327)	(0.349)	(0.352)	(0.341)	(0.331)	(0.343)	(0.306)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.340	0.312	0.464	0.224	0.196	0.305	0.178	0.152	0.251
Obs.	2137	2156	2132	2145	2157	2143	2150	2140	2129
Panel B: Time to maturity									
		Short			**Middle**			**Long**	
$CallOIB_t$	0.077***		0.050***	0.058***		0.046***	0.036***		0.033***
	(0.004)		(0.004)	(0.003)		(0.003)	(0.003)		(0.003)
$CallOIB_{t-1}$	−0.038***		−0.013**	−0.024***		−0.018***	−0.018***		−0.014***
	(0.005)		(0.005)	(0.003)		(0.003)	(0.004)		(0.004)
$PutOIB_t$		−0.075***	−0.053***		−0.046***	−0.033***		−0.024***	−0.019***
		(0.004)	(0.004)		(0.004)	(0.003)		(0.003)	(0.003)
$PutOIB_{t-1}$		0.036***	0.017***		0.014***	0.006***		0.011***	0.005**
		(0.005)	(0.005)		(0.003)	(0.002)		(0.003)	(0.002)
$Return_{t-1}$	0.094	0.045	0.076	0.082	−0.006	0.069	0.074	0.028	0.083
	(0.065)	(0.057)	(0.064)	(0.053)	(0.059)	(0.056)	(0.086)	(0.086)	(0.094)
Intercept	0.217	0.249	0.115	0.145	0.073	0.220	0.486	0.588	0.531
	(0.369)	(0.391)	(0.354)	(0.349)	(0.354)	(0.339)	(0.532)	(0.551)	(0.514)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.309	0.325	0.418	0.255	0.215	0.351	0.187	0.126	0.240
Obs.	1552	1556	1552	1919	1919	1919	952	952	952
Panel C: Trade size									
		Small			**Medium**			**Large**	
$CallOIB_t$	0.027***		0.032***	0.056***		0.048***	0.074***		0.050***
	(0.004)		(0.004)	(0.004)		(0.004)	(0.003)		(0.003)
$CallOIB_{t-1}$	−0.020***		−0.012***	−0.035***		−0.017***	−0.038***		−0.016***
	(0.003)		(0.003)	(0.004)		(0.004)	(0.004)		(0.004)
$PutOIB_t$		−0.031***	−0.034***		−0.047***	−0.039***		−0.068***	−0.045***
		(0.004)	(0.004)		(0.003)	(0.003)		(0.003)	(0.003)
$PutOIB_{t-1}$		0.013***	0.008***		0.016***	0.004		0.032***	0.011***

(Continues)

TABLE 4 | (Continued)

Panel C: Trade size

	Small			**Medium**			**Large**		
		(0.003)	(0.002)		(0.003)	(0.003)		(0.004)	(0.004)
Return_{t-1}	0.012	0.006	0.024	0.034	0.032	0.046	0.126**	0.073	0.101*
	(0.055)	(0.052)	(0.053)	(0.051)	(0.048)	(0.048)	(0.053)	(0.054)	(0.056)
Intercept	0.242	−0.210	0.158	0.790**	−0.416	0.169	0.552	0.651*	0.411
	(0.377)	(0.364)	(0.360)	(0.356)	(0.343)	(0.313)	(0.341)	(0.354)	(0.356)
Controls	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.054	0.079	0.134	0.151	0.159	0.254	0.308	0.307	0.410
Obs.	2161	2161	2161	2161	2161	2161	2161	2161	2161

Note: Panel A, B, and C of the table report the regression results,

$\text{Return}_t = \alpha + \beta_0\text{CallOIB}_t + \beta_1\text{CallOIB}_{t-1} + \gamma_0\text{PutOIB}_t + \gamma_1\text{PutOIB}_{t-1} + \lambda\text{Controls}_{t-1} + \phi\text{Return}_{t-1} + \varepsilon_t,$

for different subsamples based option moneyness, time to maturity and option trade size, respectively. In the regression, Return_t is the open-to-close return of the SSE 50 ETF on day t. CallOIB_t and PutOIB_t represent call and put option order imbalance on day t, respectively. The control variables include TermSpread, MarketHV, Turnover, ILLIQ, PCR, OSR, IVS, SKEW, POV and NOV. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

TABLE 5 | Changes in implied volatility and option order imbalance.

	OTM		**ATM**		**ITM**	
	$\Delta CallIV_t^D$	$\Delta PutIV_t^D$	$\Delta CallIV_t^D$	$\Delta PutIV_t^D$	$\Delta CallIV_t^D$	$\Delta PutIV_t^D$
CallOIB_t^D	0.029***	0.005	0.017**	0.007	0.003	0.002
	(0.010)	(0.005)	(0.008)	(0.010)	(0.006)	(0.017)
PutOIB_t^D	0.001	0.016**	0.003	0.003	0.002	−0.016
	(0.008)	(0.006)	(0.006)	(0.007)	(0.005)	(0.010)
$\Delta\text{CallIV}_{t-1}^D$	−0.120**		−0.098		−0.270***	
	(0.058)		(0.072)		(0.040)	
$\Delta\text{PutIV}_{t-1}^D$		−0.154**		−0.160*		−0.213***
		(0.077)		(0.082)		(0.073)
Return_t	−0.567***	0.065	−0.343*	−0.376**	−0.300*	−0.109
	(0.207)	(0.111)	(0.180)	(0.178)	(0.154)	(0.277)
Amount_t	0.001	0.002***	0.001	0.003***	0.000	0.004***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Intercept	0.030	−0.268***	0.170	−0.337*	0.003	−0.853**
	(0.190)	(0.095)	(0.186)	(0.185)	(0.193)	(0.417)
Adj. R^2	0.072	0.044	0.036	0.060	0.084	0.054
Obs.	2399	2401	2416	2416	2400	2398

Note: This table reports the results of the following regressions:

$$\Delta\text{CallIV}_t^D = \alpha + \beta_1\text{CallOIB}_t^D + \beta_2\text{PutOIB}_t^D + \theta\Delta\text{CallIV}_{t-1}^D + \phi_1\text{Return}_t + \phi_2\text{Amount}_t + \varepsilon_t$$

$$\Delta\text{PutIV}_t^D = \alpha + \beta_1\text{CallOIB}_t^D + \beta_2\text{PutOIB}_t^D + \theta\Delta\text{PutIV}_{t-1}^D + \phi_1\text{Return}_t + \phi_2\text{Amount}_t + \varepsilon_t$$

where D = ITM, ATM, OTM, representing different option moneyness categories. ΔCallIV_t^D and ΔPutIV_t^D represent the changes in call and put option implied volatility within option category D on day t, respectively. CallOIB_t^D and PutOIB_t^D denote the call and put order imbalance within option category D, respectively. And Return_t and Amount_t represent the open-to-close return and trading amount of underlying ETF, respectively. All variables except Amount_t are expressed in percentage terms, and Amount_t is expressed in units of ten million. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

In contrast, under the limits-to-arbitrage hypothesis, order flow induces inventory pressure on market makers, compelling them to adjust prices to manage inventory risk. However, as market makers eventually rebalance their positions, option prices—and thus implied volatilities—will partially revert to prior levels. Consequently, this mechanism predicts that changes in implied volatility will exhibit significant negative autocorrelation, consistent with transitory, rather than permanent, price effects.

Furthermore, the implications for cross-option effects differ between these two channels. If order flow impacts the IVF primarily through informed trading, both call and put order flows should affect implied volatility symmetrically, reflecting the put-call parity and the theoretical equivalence between calls and puts. In contrast, if inventory pressure predominates, the impact should be type-specific: call (put) order flows should primarily affect the implied volatility of calls (puts), rather than both.

Inspired by Bollen and Whaley (2004), we estimate the following regressions:

$$\begin{aligned}\Delta \mathrm{CallIV}_t^D = \alpha &+ \beta \mathrm{CallOIB}_t^D + \gamma \mathrm{PutOIB}_t^D \\ &+ \theta \Delta \mathrm{CallIV}_{t-1}^D + \phi_1 \mathrm{Return}_t \ , \\ &+ \phi_2 \mathrm{Amount}_t + \varepsilon_t,\end{aligned} \tag{8}$$

$$\begin{aligned}\Delta \mathrm{PutIV}_t^D = \alpha &+ \beta \mathrm{CallOIB}_t^D + \gamma \mathrm{PutOIB}_t^D \\ &+ \theta \Delta \mathrm{PutIV}_{t-1}^D + \phi_1 \mathrm{Return}_t \ , \\ &+ \phi_2 \mathrm{Amount}_t + \varepsilon_t,\end{aligned} \tag{9}$$

where $D = \{\mathrm{OTM}, \mathrm{ATM}, \mathrm{ITM}\}$. In the regressions above, $\Delta \mathrm{CallIV}_t^D$ and $\Delta \mathrm{PutIV}_t^D$ represent the changes in the implied volatility of calls and puts within option series D on day t, respectively. For example, $\Delta \mathrm{CallIV}_t^{\mathrm{OTM}} = \mathrm{CallIV}_t^{\mathrm{OTM}} - \mathrm{CallIV}_{t-1}^{\mathrm{OTM}}$, where $\mathrm{CallIV}_t^{\mathrm{OTM}}$ is the volume-weighted average implied volatility of OTM call options on day t. Return_t and Amount_t denote the return and trading amount of underlying, respectively.

Based on the above arguments, we test the following hypotheses:

H1a. *Under the inventory pressure channel, there is a significant negative autocorrelation in changes in implied volatility for both call and put options; specifically, $\theta < 0$ in Equations (8) and (9).*

H1b. *Under the inventory pressure channel, the order imbalance of call (put) options only affects the implied volatility changes of call (put) options. Specifically, in Equation (8) (for calls), $\beta \neq 0$ and $\gamma = 0$; in Equation (9) (for puts), $\beta = 0$ and $\gamma \neq 0$.*

Table 5 presents the test results, with a focus on out-of-the-money (OTM) options for illustrative purposes. In Columns 1 and 2, the estimated values of θ are −0.120 and −0.154, respectively, both of which are statistically significant at the 5% level. This finding supports hypothesis H1a, confirming the presence of significant negative autocorrelation in changes in implied volatility for both call and put options. Furthermore, in Column 1, the coefficient β is 0.029 and statistically significant at the 1% level, while γ is 0.001 and not statistically significant. In Column 2, β is 0.005 and not significant, whereas γ is 0.016 and statistically significant at the 5% level. These results are consistent with H1b, indicating that the order imbalance of call (put) options primarily affects the implied volatility changes of call (put) options, but not the opposite. Overall, these findings suggest that order imbalance in OTM options mainly reflects

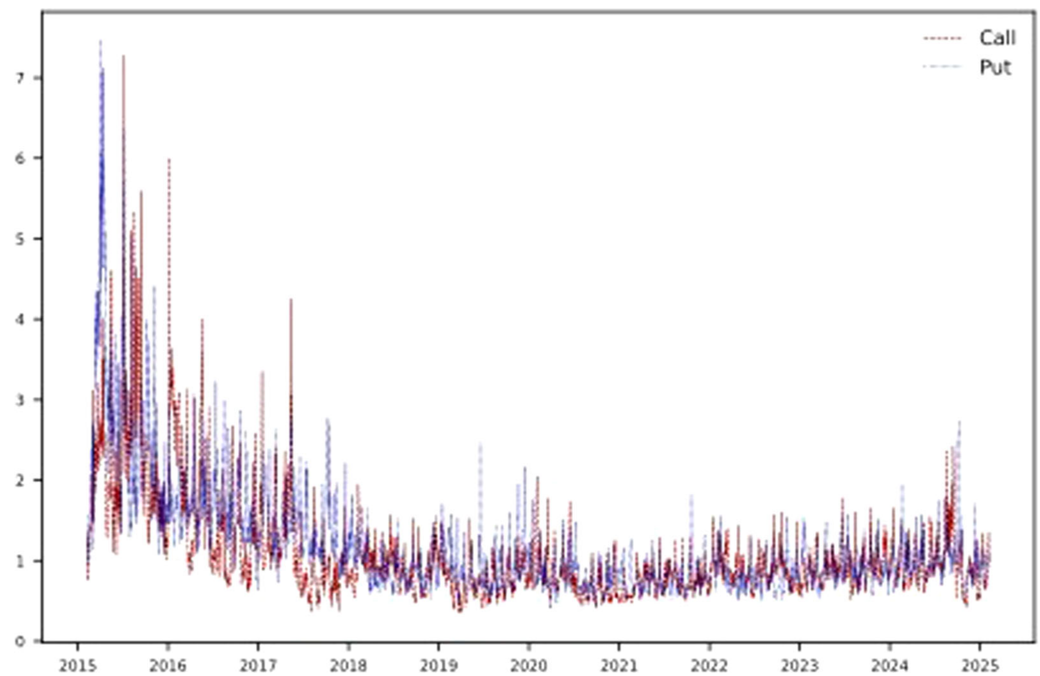

FIGURE 3 | Daily time series of the volume-weighted average of option bid-ask spreads. The red dashed line and blue dotted line represent the bid-ask spreads of calls and puts, respectively.

inventory pressure faced by market makers, rather than informed trading by market participants.

For at-the-money (ATM) and in-the-money (ITM) options, although H1a and H1b are not simultaneously supported, at least one hypothesis is confirmed in each case, and the signs of the estimated coefficients remain consistent with theoretical expectations. This outcome may reflect the fact that OTM options are the most actively traded, rendering the results for this subsample more robust.

In summary, following the framework of Bollen and Whaley (2004), our findings provide evidence against the notion that option order imbalance is primarily driven by informed trading activity. Furthermore, as shown in Appendix Table A3, our results remain robust across different option maturities.

4.4.2 | Return Reversal and Option Liquidity

In theory, informed traders are more likely to engage in options trading when market liquidity is high, as ample liquidity facilitates the execution of large trades with lower transaction costs. In contrast, when option liquidity is low, it becomes more challenging for market makers to unwind their positions, thereby amplifying inventory-related hedging pressure. This setting allows us to further distinguish between

TABLE 6 | Return reversal under low option liquidity.

	High bid-ask spreads			High absolute ILLIQ		
	(1)	(2)	(3)	(4)	(5)	(6)
CallOIB_t	0.077***		0.050***	0.080***		0.052***
	(0.003)		(0.004)	(0.003)		(0.004)
CallOIB_{t-1}	−0.037***		−0.015***	−0.039***		−0.016***
	(0.004)		(0.004)	(0.004)		(0.004)
$\text{CallOIB}_t{*}D_t^{\text{Call}}$	0.017***		0.015***	0.010***		0.010***
	(0.003)		(0.003)	(0.003)		(0.003)
$\text{CallOIB}_{t-1}{*}D_{t-1}^{\text{Call}}$	−0.008***		−0.008***	−0.005**		−0.004*
	(0.003)		(0.003)	(0.003)		(0.002)
PutOIB_t		−0.072***	−0.047***		−0.073***	−0.046***
		(0.004)	(0.004)		(0.004)	(0.004)
PutOIB_{t-1}		0.027***	0.007		0.030***	0.008*
		(0.004)	(0.005)		(0.004)	(0.005)
$\text{PutOIB}_t{*}D_t^{\text{Put}}$		−0.010***	−0.005		−0.009**	−0.008**
		(0.004)	(0.004)		(0.004)	(0.003)
$\text{PutOIB}_{t-1}{*}D_{t-1}^{\text{Put}}$		0.011***	0.007*		0.004	0.005
		(0.004)	(0.004)		(0.004)	(0.004)
Return_{t-1}	0.110**	0.068	0.089*	0.114**	0.075	0.098*
	(0.050)	(0.047)	(0.051)	(0.049)	(0.047)	(0.051)
Intercept	0.836***	0.565*	0.503	0.924***	0.543*	0.590*
	(0.310)	(0.330)	(0.310)	(0.313)	(0.330)	(0.314)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.331	0.323	0.429	0.326	0.321	0.426
Obs.	2396	2396	2396	2396	2396	2396

Note: The table reports the results of the following regressions:

$$\begin{aligned}\text{Return}_t = \alpha &+ \beta_0 \text{CallOIB}_t + \beta_1 \text{CallOIB}_{t-1} + \beta_0' \text{CallOIB}_t{*}D_t^{\text{Call}} \\ &+ \beta_1' \text{CallOIB}_{t-1}{*}D_{t-1}^{\text{Call}} + \gamma_0 \text{PutOIB}_t + \gamma_1 \text{PutOIB}_{t-1} + \gamma_0' \text{PutOIB}_t \\ &{*}D_t^{\text{Put}} + \gamma_1' PutOIB_{t-1}{*}D_{t-1}^{\text{Put}} + \lambda \text{Control}_{t-1} + \phi \text{Return}_{t-1} + \varepsilon_t,\end{aligned}$$

where $D_t^{\text{Call}} = 1$ if the liquidity measure of calls on day t is greater than its' average over past 30 days, and 0 otherwise. Similarly, $D_t^{Put} = 1$ if the liquidity measure of puts on day t is greater than its' average over past 30 days, and 0 otherwise. In Column 1–3 and 4–6, the option liquidity measures are relative bid-ask spreads and absolute ILLIQ, respectively. In the regression, Return_t is the open-to-close return of the SSE 50 ETF on day t. CallOIB_t and PutOIB_t represent call and put option order imbalance on day t, respectively. The control variables include TermSpread, MarketHV, Turnover, ILLIQ, PCR, **OSR**, IVS, SKEW, POV, and NOV. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

hedging-induced price pressure and informed trading activity. Specifically, we propose the following hypothesis:

H2. *The return reversal phenomenon is stronger when option market liquidity is lower.*

To test this hypothesis, we construct an option liquidity measure based on the relative bid-ask spread. Given that trading in calls and puts may reflect different directional bets, we compute bid-ask spreads for call and put options separately. As shown in Figure 3, both call and put option bid-ask spreads exhibit considerable time variation; notably, spreads are wider in the early part of the sample and narrow over time. Based on this pattern, we define periods of high call (put) bid-ask spreads as those in which the current spread exceeds its 30-day moving average.

The empirical results, reported in Column 1 of Table 6, reveal that the return reversal effect associated with call order imbalance is significantly more pronounced during periods of high call bid-ask spreads. For instance, the contemporaneous effect is 0.077 in normal liquidity periods, rising by 0.017 in high spread periods. Similarly, the reversal effect is −0.037 in normal periods, becoming even more negative by −0.008 during high spread periods. These findings indicate that, during times of reduced call option liquidity, inventory pressure arising from call order imbalance exerts a stronger influence on underlying prices: the initial positive price impact becomes larger, and the subsequent reversal is sharper.

Column 2 reveals symmetric evidence for put order imbalance: during periods of high put option bid-ask spreads, the contemporaneous effect declines further and the lagged effect increases more, resulting in a larger overall price impact compared to normal periods. Columns 4–6 report results using absolute ILLIQ[9] as an alternative liquidity measure following Cao and Wei (2010), and the conclusions remain robust, further supporting Hypothesis H2.

In summary, taken together with previous evidence, our results indicate that the return reversal between option order imbalance and underlying returns is primarily driven by market makers' hedging activity rather than informed trading.

5 | Further Analysis

In this section, we present several additional empirical findings. First, we examine how the deregulation event in China's derivatives markets have influenced options trading, and analyze whether the magnitude of the return reversal effect has diminished over time through subperiod analysis. Second, we evaluate the out-of-sample forecasting ability of our results and discuss the construction of a profitable trading strategy based on the documented phenomenon. Third, we explore how market makers' rebalancing activities affect the realized volatility of the underlying asset. Finally, we extend our empirical analysis to two additional ETFs with listed options on the SSE, utilizing panel data methods.

5.1 | Event Study and Subperiod Analysis

We begin by investigating the impact of changes in index futures trading restrictions on the options market. Between August and September 2015, the China Securities Regulatory Commission (CSRC) imposed stricter restrictions on index futures trading, including higher margin requirements, increased transaction fees, and tighter position limits (e.g., a daily cap of 10 non-hedging opening trades). Starting December 3, 2018,

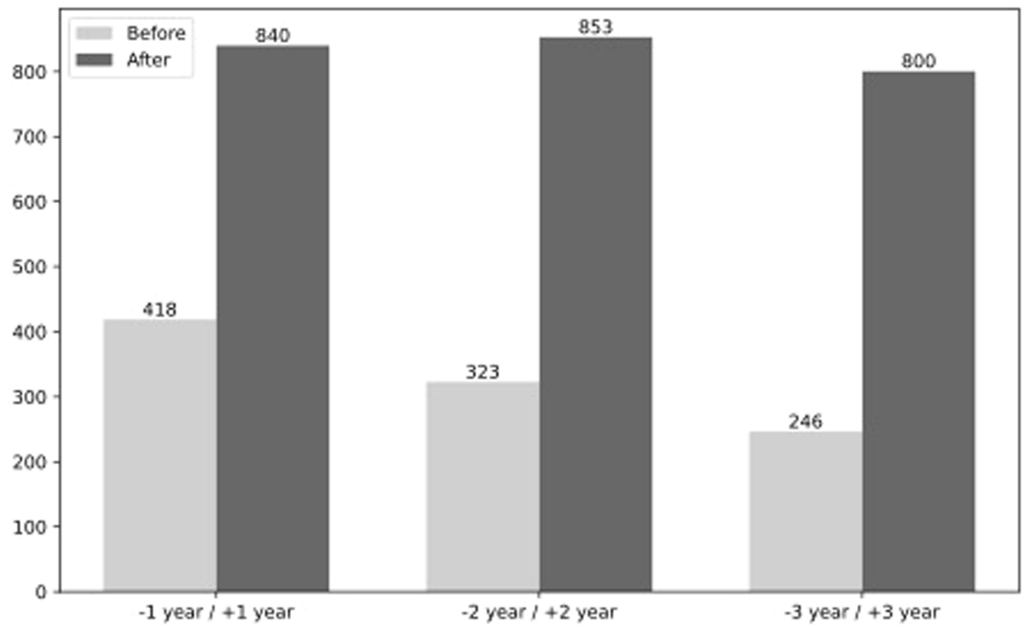

FIGURE 4 | Average daily option trading volume within event windows before and after the event. The event refers to the relaxation of restrictions on index futures on December 3, 2018. In the graph, from left to right, the bars represent the average daily option trading volume within 1 year, 2 years, and 3 years before and after the event, respectively.

these restrictions were gradually relaxed, eventually increasing the cap for nonhedging opening trades to 200 contracts. Although these regulatory changes primarily targeted the futures market, we anticipate that they also influenced participation and activity in the options market.

Figure 4 shows the average daily options trading volume surrounding the deregulation on December 3, 2018. In the year preceding the event, average daily volume was 418,000 contracts, compared to 840,000 contracts in the year after. This increase exceeds what would be expected from the general early-stage growth of the market: in the 2 years before deregulation, average daily volume was 323,000 contracts, rising to 853,000 contracts in the 2 years after. These figures suggest that the easing of index futures trading restrictions significantly boosted options trading activity, potentially increasing inventory pressure on option market makers and thereby intensifying the return reversal effect.

Following Luo et al. (2020), we define the event window as the 3 years before and after December 3, 2018, and construct a dummy variable E_t equal to 1 in the post-event period and 0 in the pre-event period. The results of Table 7 present the return reversal results before and after deregulation. For call order imbalance, the contemporaneous effect increases from 0.074 pre-event to 0.100 (0.074 + 0.026) postevent, while the reversal effect becomes more pronounced, moving from −0.031 to −0.056 (−0.031 + (−0.025)). This provides clear evidence that the magnitude of the return reversal effect intensified following the policy change.

To further assess the robustness of the return reversal phenomenon and investigate whether its magnitude has diminished in recent years, we estimate the return reversal effect over two subperiods: February 9, 2018 to August 8, 2021, and August 9, 2021 to February 9, 2025. The results in Table 8 show that the return reversal effect has indeed weakened in the more recent subperiods. For example, in Columns 1 and 3, the return reversal associated with call order imbalance are −0.050 and −0.036, respectively, whereas the corresponding effects in Columns 4 and 6 are −0.019 and −0.014. This attenuation is consistent with the proposition by Chordia et al. (2014) that the return reversal effect tends to decline as market liquidity increases.[10]

TABLE 7 | Relaxing the restrictions on Index futures.

	(1)	(2)	(3)
CallOIB_t	0.074***		0.050***
	(0.004)		(0.004)
CallOIB_{t-1}	−0.031***		−0.010**
	(0.005)		(0.004)
$\text{CallOIB}_t{}^*E_t$	0.026***		0.020***
	(0.006)		(0.007)
$\text{CallOIB}_{t-1}{}^*E_t$	−0.025***		−0.016**
	(0.006)		(0.007)
PutOIB_t		−0.071***	−0.046***
		(0.005)	(0.005)
PutOIB_{t-1}		0.023***	0.003
		(0.004)	(0.004)
$\text{PutOIB}_t{}^*E_t$		−0.011*	−0.006
		(0.006)	(0.007)
$\text{PutOIB}_{t-1}{}^*E_t$		0.010*	−0.001
		(0.005)	(0.006)
Return_{t-1}	−0.003	0.007	−0.006
	(0.044)	(0.035)	(0.038)
Intercept	0.260	1.108**	0.298
	(0.549)	(0.490)	(0.486)
Controls	Yes	Yes	Yes
Adj. R^2	0.358	0.351	0.472
Obs.	1462	1462	1462

Note: The table reports the results of the following regression:

$$\text{Return}_t = \alpha + \beta_0\text{CallOIB}_t + \beta_1\text{CallOIB}_{t-1} + \beta_0'\text{CallOIB}_t{}^*E_t + \beta_1'\text{CallOIB}_{t-1}{}^*E_t + \gamma_0\text{PutOIB}_t + \gamma_1\text{PutOIB}_{t-1} + \gamma_0'\text{PutOIB}_t{}^*E_t + \gamma_1'\text{PutOIB}_{t-1}{}^*E_t + \lambda\text{Control}_{t-1} + \phi\text{Return}_{t-1} + \varepsilon_t.$$

The sample period is from December 3, 2015, to December 3, 2021. $E_t = 1$ if t is later than December 3, 2018, and 0 otherwise. In the regression, Return_t is the open-to-close return of the SSE 50 ETF on day t. CallOIB_t and PutOIB_t represent call and put option order imbalance on day t, respectively. The control variables include TermSpread, MarketHV, Turnover, ILLIQ, PCR, OSR, IVS, SKEW, POV, and NOV. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively.

5.2 | Out-of-Sample Analysis and Trading Strategy Performance

The preceding in-sample analysis provides convincing and robust evidence for a return reversal effect. In this subsection, we further validate our findings through an out-of-sample analysis. We continue to define Return_t as the open-to-close return of the underlying asset when conducting the out-of-sample forecast and constructing the trading strategy. However, all of our conclusions remain unchanged when close-to-close returns are used instead, given that the ETF still operates under a T+1 trading mechanism.

To evaluate the out-of-sample predictability of underlying returns, we adopt a recursive forecasting approach. The full sample period is divided into two segments: the initial training period, from February 9, 2015, to February 8, 2022, and the testing (out-of-sample) period, from February 9, 2022, to February 9, 2025.

In the first step, we estimate the following regression model using the training sample, which includes all observations from day 1 to day t:

$$\begin{aligned}\text{Return}_t = \alpha &+ \beta_0\text{CallOIB}_t + \beta_1\text{CallOIB}_{t-1} \\ &+ \gamma_0\text{PutOIB}_t + \gamma_1\text{PutOIB}_{t-1} \\ &+ \lambda\text{Controls}_{t-1} + \phi\text{Return}_{t-1} + \varepsilon_t\end{aligned} \quad . \qquad (10)$$

TABLE 8 | Open-to-close return and option order imbalance within different periods.

	First subperiod: 2018.2.9–2021.8.8			Second subperiod:2021.8.9–2025.2.9		
	(1)	(2)	(3)	(4)	(5)	(6)
CallOIB_t	0.102***		0.080***	0.084***		0.069***
	(0.005)		(0.005)	(0.003)		(0.004)
CallOIB_{t-1}	−0.050***		−0.036***	−0.019***		−0.014***
	(0.006)		(0.005)	(0.005)		(0.004)
PutOIB_t		−0.079***	−0.043***		−0.062***	−0.028***
		(0.004)	(0.003)		(0.003)	(0.003)
PutOIB_{t-1}		0.024***	0.005		0.026***	0.009***
		(0.005)	(0.005)		(0.004)	(0.003)
Return_{t-1}	−0.039	0.017	-0.013	-0.002	0.016	0.006
	(0.034)	(0.028)	(0.031)	(0.035)	(0.042)	(0.035)
Intercept	0.406	0.746	0.690	0.758**	0.595*	0.896***
	(0.531)	(0.565)	(0.518)	(0.300)	(0.360)	(0.296)
Controls	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.449	0.323	0.526	0.522	0.319	0.570
Obs.	847	847	847	845	845	845

Note: The table report the regression results,

$$\text{Return}_t = \alpha + \beta_0 \text{CallOIB}_t + \beta_1 \text{CallOIB}_{t-1} + \gamma_0 \text{PutOIB}_t + \gamma_1 \text{PutOIB}_{t-1} + \lambda \text{Controls}_{t-1} + \phi \text{Return}_{t-1} + \varepsilon_t,$$

for different subperiods. In the regression, Return_t is the open-to-close return of the SSE 50 ETF on day t. CallOIB_t and PutOIB_t represent call and put option order imbalance on day t, respectively. The control variables include TermSpread, MarketHV, Turnover, ILLIQ, PCR, OSR, IVS, SKEW, POV and NOV. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2018, to February 9, 2025.

In the second step, we calculate the one-step-ahead out-of-sample forecast of the underlying return $\hat{r}_{t+1}$, by substituting actual CallOIB_{t+1}, PutOIB_{t+1}, and other variables observed at period t into the estimated model. Note that CallOIB_{t+1} and PutOIB_{t+1} are not available at period t, we assume for illustration that they are obtainable; they will later be replaced by predicted values. These two steps are repeated recursively, with each new observation added to the training sample.

Out-of-sample predictive power is evaluated using the out-of-sample R^2, R^2_{OS}, following Campbell and Thompson (2007):

$$R^2_{\text{OS}} = 1 - \frac{\sum(\text{Return}_s - \hat{r}_s)^2}{\sum(\text{Return}_s - \bar{r}_s)^2}, \tag{11}$$

where $\hat{r}_s$ denotes the one-step-ahead forecast and Return_s represents is the actual return. And the benchmark forecast, $\bar{r}_s$, is the historical average of the underlying returns. A positive value of R^2_{OS} indicates that the forecasting model in Equation (11) produces smaller errors than the historical average. Columns 1–3 of Panel A in Table 9 show that all R^2_{OS} values are greater than zero, and their magnitudes are substantially larger than those documented in prior studies that use SSE 50 ETF option-implied variables for forecasting.

We next construct a trading strategy based on the out-of-sample forecast results. Specifically, if the forecasted return $\hat{r}_{t+1}$ on day t is positive, the strategy buys one share of the underlying ETF at the open price on day t+1 and sells it at the closing price. If $\hat{r}_{t+1}$ is less than or equal to zero, no trade is executed. The return of this strategy is calculated as follows:

$$\text{Return}_s^{\text{trading}} = \begin{cases} \text{Return}_s, & \text{if} \quad \hat{r}_s > 0 \\ 0, & \text{if} \quad \hat{r}_s \leq 0 \end{cases}. \tag{12}$$

Columns 4–6 of Panel A present the performance of this trading strategy. Notably, in Column 6, over 726 out-of-sample trading days, the strategy executed buy trades on 327 days, achieving an average gain of 0.63%.

However, the procedure described above is not feasible in practice, as it relies on unavailable future values for out-of-sample forecasting. To address this, we propose an alternative, implementable approach: in the second step of the recursive forecasting procedure, we replace the unavailable CallOIB_{t+1} and PutOIB_{t+1}, with their predicted values generated from AR(1) models. These predicted order imbalances and other available variables are then substituted into the regression model estimated in the first step to generate the out-of-sample forecast, $\hat{r}_s$. All other procedures remain as previously described. Panel B of Table 9 presents the results of this practical approach. Not surprisingly, both the out-of-sample forecasting performance and the returns from the associated trading strategies are substantially lower than those reported in Panel A, which uses actual order imbalance.

TABLE 9 | Recursively out-of-sample R^2 and trading strategy performance.

Panel A: Based on unavailable order imbalance						
	Out of sample R^2			Average return of trading strategy		
	Model 1	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
call	32.77			0.67***(0.06)		
put		28.81			0.47***(0.04)	
call and put			41.28			0.63***(0.05)

Panel B: Based on predicted order imbalance						
	Out of sample R^2			Average return of trading strategy		
	Model 1	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
call	−6.36			−0.07(0.05)		
put		−3.92			−0.05(0.06)	
call and put			−6.10			−0.03(0.06)

Note: The table reports the recursively out-of-sample R^2 and trade strategy returns. In Panel A, the out-of-sample forecasting procedure consists of two steps. Step 1, estimating the regression model:

$$\text{Return}_t = \alpha + \beta_0 \text{CallOIB}_t + \beta_1 \text{CallOIB}_{t-1} + \gamma_0 \text{PutOIB}_t + \gamma_1 \text{PutOIB}_{t-1} + \lambda \text{Controls}_{t-1} + \phi \text{Return}_{t-1} + \varepsilon_t,$$

using data from the training period, 1, ..., t. Step 2, Obtaining the predicted return for period $t+1$, $\hat{r}_{t+1}$, by substituting the observable variables at period t and the unavailable variables (CallOIB_{t+1}, PutOIB_{t+1}) into the model. The control variables include TermSpread, MarketHV, Turnover, ILLIQ, PCR, OSR, IVS, SKEW, POV, and NOV. The out-of-sample R^2 is computed as

$$R^2_{\text{OS}} = 1 - \frac{\sum(\text{Return}_s - \hat{r}_s)^2}{\sum(\text{Return}_s - \bar{r}_s)^2},$$

where Return_s is the actual daily open-to-close return of underlying in testing period s, and $\hat{r}_s$ and $\bar{r}_s$ are the out-of-sample forecasted return and the historical average return, respectively. The trading strategy is that if the forecasted return $\hat{r}_s > 0$, buy one share of the underlying at the open price and sell it at the close price. The strategy return is defined as,

$$\text{Return}_s^{\text{trading}} = \begin{cases} \text{Return}_s, & \text{if} \quad \hat{r}_s > 0 \\ 0, & \text{if} \quad \hat{r}_s \le 0 \end{cases}.$$

In Panel B, the procedure is similar to that in the Panel A, except that in the step 2, to obtain the predicted return for period $t+1$ in practice, the unavailable variables are replaced by their AR(1) predicted value during period t. Model 1 and Model 2 represent that in step 1 the regression model contains call and put order imbalance separately, and Model 3 represent that the model contains call and put order imbalance simultaneously. The full sample period is from February 9, 2015, to February 9, 2025, and is divided into two segments. The period from February 9, 2015, to February 8, 2022, serves as the initial training period. The period from February 9, 2022, to February 9, 2025, constitutes the out-of-sample testing period. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively.

It is worth noting that, under the market makers' inventory pressure channel, the predictive relationship appears to be very short-term and not persistent, which may explain why out-of-sample predictability is weak.[11] This also suggests, from another perspective, that the Chinese options market achieved a relatively high level of efficiency during our sample period.

5.3 | The Stabilized Effect on Underlying Returns

In this section, we examine the effect of market makers' dynamic hedging on the volatility of the underlying asset. As discussed in Barbon and Buraschi (2021), the net gamma exposure on day t may be influenced by contemporaneous movements in the underlying price. To address this potential endogeneity, we follow Barbon and Buraschi (2021) and employ the lagged value of net gamma exposure as an instrumental variable in the following regression:

$$\text{RV}_t = \alpha + \beta_1 \text{NGE}_{t-1} + \beta_2 \text{NGE}_{t-2} + \gamma \text{Controls}_{t-1} + \rho \text{RV}_{t-1} + \varepsilon_t, \tag{13}$$

where RV_t is the realized volatility of the underlying on day t and NGE_{t-1} is the net gamma exposure on day $t-1$. The control variables are consistent with those used in Section 4. Our results, presented in Column 1 of Table 10, indicate that underlying volatility is negatively related to net gamma exposure. This negative relationship remains robust even after accounting for volatility clustering, as shown in Columns 2 and 3. When incorporating the first two lags of net gamma exposure (Columns 4 and 5), we find that only the first lag is statistically significant, suggesting that the effect of net gamma exposure on volatility is short-lived. This finding aligns with the theory that market makers' dynamic hedging demand is transitory. Overall, our results suggest that market makers' dynamic hedging activities contribute to the stabilization of the underlying market.

TABLE 10 | Spot realized volatility and option net gamma exposure.

	(1)	(2)	(3)	(4)	(5)
NGE_{t-1}	−0.010***	−0.002***	−0.002***	−0.002***	−0.001***
	(0.001)	(0.000)	(0.000)	(0.001)	(0.001)
NGE_{t-2}				0.000	0.000
				(0.001)	(0.000)
RV_{t-1}		0.749***	0.548***	0.548***	0.381***
		(0.049)	(0.054)	(0.054)	(0.048)
RV_{t-2}			0.267***	0.268***	0.183***
			(0.035)	(0.035)	(0.036)
$TermSpread_{t-1}$					0.088
					(0.268)
$MarketHV_{t-1}$					0.007***
					(0.002)
$Turnover_{t-1}$					0.031***
					(0.008)
$ILLIQ_{t-1}$					0.000
					(0.002)
PCR_{t-1}					0.003*
					(0.002)
OSR_{t-1}					0.000
					(0.000)
IVS_{t-1}					−0.007**
					(0.003)
$SKEW_{t-1}$					−0.015***
					(0.005)
POV_{t-1}					−0.003***
					(0.001)
NOV_{t-1}					0.003***
					(0.001)
Intercept	1.263***	0.317***	0.234***	0.232***	0.006
	(0.059)	(0.052)	(0.042)	(0.043)	(0.113)
Adj. R^2	0.130	0.619	0.646	0.646	0.676
Obs.	2191	2191	2190	2190	2161

Note: The table reports the results of the following regression:

$$RV_t = \alpha + \beta_1 NGE_{t-1} + \beta_2 NGE_{t-2} + \phi_1 RV_{t-1} + \phi_2 RV_{t-2} + \lambda Controls_{t-1} + \varepsilon_t,$$

where RV_t is the realized volatility of underlying on day t. NGE_{t-1} and NGE_{t-2} denote the net gamma exposure of option market makers on day $t-1$ and $t-2$, respectively. The control variables include TermSpread, MarketHV, Turnover, ILLIQ, PCR, OSR, IVS, SKEW, POV and NOV. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

5.4 | Robustness Analysis With Additional Index ETF Options

In this section, we conduct a robustness analysis to validate our previous findings by incorporating two additional ETF options listed on the SSE: the CSI 300 ETF options and the CSI 500 ETF options, which were introduced on December 23, 2019, and September 19, 2022, respectively. We aggregate the data from all three ETFs and their corresponding options and estimate the following unbalanced panel regression models:

$$\begin{aligned} Return_{i,t} = \alpha + \beta_0 CallOIB_{i,t} + \beta_1 CallOIB_{i,t-1} + \gamma_0 PutOIB_{i,t} \\ + \gamma_1 PutOIB_{i,t-1} + \lambda Controls_{i,t-1} + \phi Return_{i,t-1} + \mu_i + \varepsilon_{i,t}, \end{aligned} \tag{14}$$

TABLE 11 | Panel data regression.

Panel A: Spot return and option order imbalance

	(1)	(2)	(3)	(4)	(5)	(6)
$CallOIB_{i,t}$	0.077***		0.059***	0.082***		0.061***
	(0.003)		(0.003)	(0.003)		(0.003)
$CallOIB_{i,t-1}$	−0.036***		−0.015***	−0.036***		−0.017***
	(0.003)		(0.003)	(0.004)		(0.004)
$PutOIB_{i,t}$		−0.069***	−0.050***		−0.075***	−0.052***
		(0.003)	(0.003)		(0.003)	(0.003)
$PutOIB_{i,t-1}$		0.035***	0.013***		0.032***	0.012***
		(0.003)	(0.003)		(0.003)	(0.004)
$Return_{i,t-1}$	0.128***	0.114***	0.09***	0.099**	0.070*	0.094**
	(0.032)	(0.030)	(0.034)	(0.041)	(0.036)	(0.039)
Intercept	0.610***	−0.286***	0.313***	0.546*	0.595*	0.626**
	(0.056)	(0.043)	(0.064)	(0.305)	(0.329)	(0.300)
Controls	No	No	No	Yes	Yes	Yes
Fixed effect	Yes	Yes	Yes	Yes	Yes	Yes
Adj. R^2	0.267	0.238	0.382	0.281	0.265	0.388
Obs.	4199	4199	4199	4109	4109	4109

Panel B: Spot volatility and option net gamma exposure

	(1)	(2)	(3)
$NGE_{i,t-1}$	−0.002***	−0.003***	−0.002***
	(0.0)	(0.001)	(0.001)
$NGE_{i,t-2}$		0.001	0.001
		(0.001)	(0.001)
$RV_{i,t-1}$	0.717***	0.718***	0.447***
	(0.043)	(0.043)	(0.046)
Intercept	0.312***	0.31***	0.234*
	(0.042)	(0.042)	(0.123)
Controls	No	No	Yes
Fixed effect	Yes	Yes	Yes
Adj. R^2	0.554	0.554	0.605
Obs.	4199	4196	4109

Note: Panel A of the table reports results of the following panel regression:

$$Return_{i,t} = \alpha + \beta_0 CallOIB_{i,t} + \beta_1 CallOIB_{i,t-1} + \gamma_0 PutOIB_{i,t} + \gamma_1 PutOIB_{i,t-1} + \lambda Controls_{i,t-1} + \phi Return_{i,t-1} + \mu_i + \varepsilon_{i,t},$$

Where $Return_{i,t}$ is the return for underlying ETF i on day t. $CallOIB_{i,t}$ and $PutOIB_{i,t}$ denote the call and put order imbalance of options with underlying ETF i, respectively. Panel B reports the results of the following regression:

$$RV_{i,t} = \alpha + \beta_1 NGE_{i,t-1} + \beta_2 NGE_{i,t-2} + \lambda Controls_{i,t-1} + \phi RV_{i,t-1} + \mu_i + \varepsilon_{i,t},$$

Where $RV_{i,t}$ is the realized volatility of ETF i on day t. $NGE_{i,t-1}$ denotes the net gamma exposure of option market makers for ETF i. The control variables, $\lambda Controls_{i,t-1}$, are defined as follows:

$$\lambda Controls_{i,t-1} = \lambda_1 TermSpread_{t-1} + \lambda_2 MarketHV_{t-1} + \lambda_3 Turnover_{i,t-1} + \lambda_4 ILLIQ_{i,t-1} + \lambda_5 PCR_{i,t-1} + \lambda_6 OSR_{i,t-1} + \lambda_7 IVS_{i,t-1} + \lambda_8 SKEW_{i,t-1} + \lambda_9 POV_{i,t-1} + \lambda_{10} NOV_{i,t-1}.$$

These panel regression model also includes the ETF fixed effects. Standard errors in parentheses are clustered at the ETF level and the Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively.

$$\mathrm{RV}_{i,t} = \alpha + \beta_1 \mathrm{NGE}_{i,t-1} + \beta_2 \mathrm{NGE}_{i,t-2} + \lambda \mathrm{Controls}_{i,t-1} + \phi \mathrm{RV}_{i,t-1} + \mu_i + \varepsilon_{i,t}. \quad (15)$$

Here $\mathrm{Return}_{i,t}$ and $\mathrm{RV}_{i,t}$ represent the daily return and realized volatility for ETF i on day t, respectively. $\mathrm{CallOIB}_{i,t}$ and $\mathrm{PutOIB}_{i,t}$ denote the call and put order imbalance of options with underlying ETF i. $\mathrm{NGE}_{i,t-1}$ denotes the net gamma exposure of option market makers for ETF i. The control variables $\mathrm{Controls}_{i,t-1}$ are consistent with those employed in Section 4, and ETF fixed effects are included to account for ETF specific characteristics.

Table 11 presents the estimation results for these two models. The panel regression results in Panel A indicate that the return reversal effect between option order imbalance and underlying returns remains significant, and the magnitude of reversal is comparable to the results obtained using the SSE 50 ETF options. Panel B demonstrates that the net gamma exposure of option market makers significantly dampens the volatility of the underlying assets. Overall, these panel regression results further reinforce our earlier conclusions and provide additional empirical support.

6 | Conclusion

In this paper, we show that order imbalances in SSE 50 ETF option calls (puts) exhibit a contemporaneous positive (negative) relationship with underlying returns, followed by a rapid reversal in the subsequent period. This feedback effect is consistent with market makers' delta-hedging activities, which generate temporary price pressure on the underlying asset. The phenomenon persists across different option categories—including moneyness, time to maturity, and trade size—and remains robust after controlling for various factors and extending the analysis to other ETF options. Additionally, we find that market makers' dynamic hedging to manage net gamma imbalances can help stabilize the underlying market. Overall, our findings contribute to the literature by providing evidence for a noninformational trading channel, highlighting how market makers' hedging behavior influences the relationship between option trading and underlying price dynamics.

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Data Availability Statement

Data are subject to third-party restrictions. The data that support the findings of this paper are available from the Chinese Stock Market and Accounting Research (https://cn.gtadata.com) and Wind Financial Terminal (www.wind.com.cn). Restrictions apply to the availability of these data, which were used under license for this study.

Endnotes

[1] For newly opened accounts, the position limit is set at 100 contracts for option rights (long positions), with a total position limit of 200 contracts and a daily buy-to-open limit of 400 contracts. As investors gain more trading experience—for example, by obtaining Level-3 trading privileges—these limits increase to 1000 contracts for option rights, 2000 contracts for total positions, and a daily buy-to-open limit of 4000 contracts. In addition to intraday trading volume limits, investors are also subject to strict purchase amount restrictions: the purchase amount may not exceed the higher of 10% of their own asset balance or 20% of the average daily market value of Shanghai and Shenzhen securities held in their securities account over the past 6 months.

[2] More specifically, primary market makers are required to consistently provide two-way quotations (bid and ask prices) at specified maximum bid-ask spreads, minimum quotation volumes, maximum response durations, minimum quote retention times, and minimum participation rates, in addition to responding to investors' quotation requests. In contrast, general market makers are only required to provide responsive quotations upon request.

[3] The CSI 300 ETF options and the CSI 500 ETF options are the cross-market option products launched on both the Shanghai Stock Exchange (SSE) and the Shenzhen Stock Exchange (SZSE). In this study, we use options listed on the SSE, with their underlying ETFs identified by the symbols 510300 and 510500, respectively.

[4] The limit to arbitrage is set to $C_t \geq \max(0, S_t - Ke^{-rT})$, $P_t \geq \max(0, Ke^{-rT} - S_t)$, where C_t is the call option price, P_t is the put option price, S_t is the closing price of underlying, K is the strike price, r is the risk-free rate, and T is the maturity date. The risk-free rate is proxied by the 3-month Shanghai Interbank Offered Rate (SHIBOR).

[5] The definitions of those two alternative weighted order imbalances are provided in the Appendix.

[6] The main conclusions of this study remain robust when using another commonly employed daily close-to-close return.

[7] Since the stationarity hypothesis is rejected, we use first differences of the variable in the regression analysis, following Chordia et al. (2021).

[8] Following Hu (2014), we classify call and put option into out-of-the-money options (OTM, $|\delta| < 0.375$), at-the-money options (ATM, $0.375 \leq |\delta| < 0.625$), and in-the-money options (ITM, $0.625 \leq |\delta|$) based on the absolute delta. With respect to time to maturity, we separate options into three categories: short (less than 30 days to maturity), middle (between 30 and 60 days to maturity) and long (greater than 60 days to maturity). We also divide all option trades into three categories based on trade size percentile within each year: small (0–30th percentile), medium (30-70th percentile), and large (70–100th percentile).

[9] The difinations of relative bid-ask spreads and absolute ILLIQ are provided in the Appendix.

[10] In contrast to the results for call order imbalance, the reversal effect of put order imbalance shows little numerical difference between Columns 2 and 4. This may be because market makers bear greater inventory risk from call options, making the dampening effect of market liquidity on return reversals more pronounced in the case of call order imbalance.

[11] In untabulated results, we extended our analysis by incorporating further lags in the AR model and by employing a VAR model to jointly predict call and put order imbalances. However, these modifications offered little improvement. Additionally, in Appendix Table A4, we report the results from a pure predictive regression approach. However, the performance is inferior to that reported in Panel A of Table 9.

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Appendix

The appendix first provides the definitions of two weighted order imbalances, the relative bid-ask spread and absolute ILLIQ, and then reports Tables A1–A4, which are not included in the main text.

Weighted order imbalance

We define the daily call and put order imbalance as the weighted summation of position-opening buyer-initiated trading volume less seller-initiated trading volume, divided by the total daily trading volume of position-opening trades, given as the following equations:

$$CallOIB_t = \frac{\sum_i w_{i,t}\left(BuyVol_{i,t}^{Call} - SellVol_{i,t}^{Call}\right)}{\sum_i\left(BuyVol_{i,t}^{Call} + SellVol_{i,t}^{Call}\right)}, \tag{1'}$$

$$PutOIB_t = \frac{\sum_j w_{j,t}\left(BuyVol_{j,t}^{Put} - SellVol_{j,t}^{Put}\right)}{\sum_j\left(BuyVol_{j,t}^{Put} + SellVol_{j,t}^{Put}\right)}. \tag{2'}$$

For the call option contract i on day t, $BuyVol_{i,t}^{Call}$ represents the trading volume initiated by the buyer, and $SellVol_{i,t}^{Call}$ represents the trading volume initiated by the seller. The numerator of the call order imbalance, $CallOIB_t$, measures the hedging demand for the underlying asset of market makers, standardized by the total daily trading volume. $PutOIB_t$ is the put order imbalance constructed similarly to $CallOIB_t$. For call delta-weighted order imbalance $w_{i,t} = \delta_{i,t}$ where $\delta_{i,t}$ is the delta of call option contract i. For call K/S-weighted order imbalance $w_{i,t} = K_{i,t}/S_t$, where $K_{i,t}$ is the strike price of call option contract i and S_t denotes the underlying price.

Relative bid-ask spreads

Specifically, for each option transaction data including trading volume, bid price and ask price, we compute the call and put bid-ask spreads as follows,

$$CallSpread_t = \sum_j Volume_{t,j}^{1\{j\in calls\}} * \frac{\mathrm{Bid}_{t,j} - Ask_{t,j}}{(\mathrm{Bid}_{t,j} + Ask_{t,j})/2} / \sum_j Volume_{t,j}^{1\{j\in calls\}}, \tag{3'}$$

$$PutSpread_t = \sum_j Volume_{t,j}^{1\{j\in puts\}} * \frac{\mathrm{Bid}_{t,j} - Ask_{t,j}}{(\mathrm{Bid}_{t,j} + Ask_{t,j})/2} / \sum_j Volume_{t,j}^{1\{j\in puts\}}, \tag{4'}$$

where $Volume_{t,j}^{1\{j\in calls\}}$ and $Volume_{t,j}^{1\{j\in puts\}}$ represent whether the jth transaction belong to calls or puts. And $Bid_{t,j}$ and $Ask_{t,j}$ denote the bid price and ask price of the jth transaction on day t, respectively.

we define the high bid-ask spread periods for call (put) options as those in which the call (put) option bid-ask spread exceeds the average spread over the past 30 days.

$$D_t^{Call} = 1\left\{CallSpread_t > \left(\frac{\sum_{j=1}^{30} CallSpread_{t-j}}{30}\right)\right\}, \tag{5'}$$

$$D_t^{Put} = 1\left\{PutSpread_t > \left(\sum_{j=1}^{30}\frac{PutSpread_{t-j}}{30}\right)\right\}. \tag{6'}$$

Absolute ILLIQ

For call option, the absolute illiquidity is defined as

$$CallAILLIQ_t = \frac{\sum_i VOL_t^i * \frac{\left|\left(P_t^i - P_{t-1}^i\right) - \delta_{t-1}^i\left(S_t - S_{t-1}\right)\right|}{DVOL_t^i}}{\sum_i VOL_t^i}, \tag{7'}$$

where VOL_t^i and $DVOL_t^i$ denote the trading volume and dollar trading volume of call option contract i on day t respectively. P_t^i, δ_{t-1}^i and S_t represent call option price, option delta and underlying price, respectively. The construction of $PutAILLIQ_t$ is analogous to that of $CallAILLIQ_t$.

Table A1 reports linear correlation coefficients across variables. Table A2 presents robustness checks using alternative underlying returns and option order imbalance specifications. Table A3 documents changes in implied volatility and order imbalance across different time-to-maturity subsamples. Table A4 presents out-of-sample predictive and trading performance based on pure forecasting regressions.

TABLE A1 | Correlation coefficients.

	Return	CallOIB	PutOIB	TermSpread	MarketHV	Turnover	ILLIQ	PCR	OSR	IVS	Skew	POV	NOV	RV	NGE
Return	1														
CallOIB	**0.49**	1													
PutOIB	**−0.48**	**−0.17**	1												
TermSpread	**0.06**	0.04	**−0.05**	1											
MarketHV	0.00	**0.20**	**0.26**	−0.01	1										
Turnover	0.00	**0.11**	**0.16**	0.02	**0.40**	1									
ILLIQ	0.04	**0.26**	**0.15**	−0.01	**0.36**	**−0.30**	1								
PCR	**−0.24**	**−0.39**	**0.29**	**−0.06**	**0.10**	**−0.10**	**0.07**	1							
OSR	−0.01	**−0.33**	**−0.38**	−0.02	**−0.55**	**−0.60**	**−0.08**	−0.01	1						
IVS	**−0.10**	**−0.26**	**−0.15**	**0.06**	**−0.52**	0.03	**−0.45**	**−0.18**	**0.24**	1					
SKEW	**0.33**	**0.27**	**−0.15**	0.02	**0.15**	**−0.20**	**0.28**	**0.23**	0.01	**−0.54**	1				
POV	**0.04**	**−0.22**	**−0.35**	0.03	**-0.16**	0.00	**−0.30**	**−0.10**	**0.62**	**0.24**	−0.02	1			
NOV	−0.03	**−0.27**	**−0.30**	0.02	**-0.16**	0.00	**−0.31**	**−0.09**	**0.62**	**0.26**	**-0.05**	**0.99**	1		
RV	0.01	**0.15**	**0.19**	-0.03	**0.63**	**0.74**	**0.05**	**−0.04**	**-0.49**	**−0.31**	−0.03	**0.06**	**0.05**	1	
NGE	**0.06**	−0.02	**-0.10**	0.00	**-0.37**	**-0.30**	**−0.10**	**−0.07**	**0.49**	**0.06**	0.02	**0.17**	**0.16**	**−0.36**	1

Note: The table reports the linear correlation coefficients, with boldface indicating significance at the 5% level.

TABLE A2 | Robustness of reversal effects.

	Close-to-close return Unweighted OIB			Open-to-close return					
				Delta-weighted OIB			K/S-weighted OIB		
$CallOIB_t$	0.085***		0.061***	0.072***		0.052***	0.077***	0.055***	
	(0.004)		(0.004)	(0.004)		(0.004)	(0.004)		(0.004)
$CallOIB_{t-1}$	−0.041***		−0.014***	−0.036***		−0.014***	−0.039***		−0.015***
	(0.004)		(0.004)	(0.004)		(0.004)	(0.004)		(0.004)
$PutOIB_t$		−0.076***	−0.052***		−0.063***	−0.043***		−0.070***	−0.047***
		(0.004)	(0.004)		(0.003)	(0.004)		(0.003)	(0.004)
$PutOIB_{t-1}$		0.034***	0.008**		0.031***	0.010**		0.035***	0.010**
		(0.003)	(0.003)		(0.004)	(0.004)		(0.004)	(0.005)
$Return_{t-1}$	0.156***	0.120***	0.075*	0.149***	0.128***	0.097**	0.154***	0.141***	0.097**
	(0.044)	(0.038)	(0.039)	(0.036)	(0.037)	(0.045)	(0.035)	(0.036)	(0.044)
Intercept	0.584***	−0.297***	0.313***	0.582***	−0.211***	0.350***	0.551***	−0.207***	0.324***
	(0.066)	(0.040)	(0.073)	(0.057)	(0.043)	(0.066)	(0.057)	(0.042)	(0.063)
Adj. R^2	0.293	0.276	0.408	0.271	0.255	0.373	0.301	0.280	0.411
Obs.	2425	2425	2425	2426	2426	2426	2426	2426	2426

Note: This table reports the results of the following regression:

$$Return_t = \alpha + \beta_0 CallOIB_t + \beta_1 CallOIB_{t-1} + \gamma_0 PutOIB_t + \gamma_1 PutOIB_{t-1} + \phi Return_{t-1} + \varepsilon_t.$$

Columns 1–3 employ daily close-to-close returns with unweighted order imbalance, while columns 4–9 use open-to-close returns. Within columns 4–9, order imbalance is calculated using delta weighting in columns 4–6 and strike-to-spot ratio (K/S) weighting in columns 7–9, where K denotes option strike price and S represents the underlying spot price. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

TABLE A3 | Changes in implied volatility and order imbalance within different time to maturity.

	Short		Middle		Long	
	$\Delta CallIV_t^D$	$\Delta PutIV_t^D$	$\Delta CallIV_t^D$	$\Delta PutIV_t^D$	$\Delta CallIV_t^D$	$\Delta PutIV_t^D$
CallOIB_t^D	0.012	0.020	0.014**	0.006	0.006	−0.005
	(0.010)	(0.013)	(0.006)	(0.007)	(0.005)	(0.007)
PutOIB_t^D	0.011	-0.003	-0.002	0.013**	0.004	0.005
	(0.011)	(0.012)	(0.005)	(0.006)	(0.005)	(0.005)
$\Delta\text{CallIV}_{t-1}^D$	−0.138**		−0.119*		−0.099*	
	(0.056)		(0.070)		(0.051)	
$\Delta\text{PutIV}_{t-1}^D$		−0.201***		−0.186***		−0.043
		(0.064)		(0.061)		(0.048)
Return_t	−0.552**	−0.563**	−0.478***	−0.200	−0.234*	0.108
	(0.226)	(0.230)	(0.174)	(0.134)	(0.133)	(0.155)
Amount_t	0.001	0.003***	0.001	0.002***	0.001	0.001***
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.000)
Intercept	0.076	−0.385	0.001	−0.197	−0.016	−0.326**
	(0.228)	(0.243)	(0.159)	(0.135)	(0.159)	(0.166)
Adj. R^2	0.071	0.080	0.074	0.069	0.037	0.019
Obs.	1857	1857	2211	2211	1138	1138

Note: This table reports the results of the following regressions:

$$\Delta\text{CallIV}_t^D = \alpha + \beta_1\text{CallOIB}_t^D + \beta_2\text{PutOIB}_t^D + \theta\Delta\text{CallIV}_{t-1}^D + \phi_1\text{Return}_t + \phi_2\text{Amount}_t + \varepsilon_t,$$

$$\Delta\text{PutIV}_t^D = \alpha + \beta_1\text{CallOIB}_t^D + \beta_2\text{PutOIB}_t^D + \theta\Delta\text{PutIV}_{t-1}^D + \phi_1\text{Return}_t + \phi_2\text{Amount}_t + \varepsilon_t.$$

where $D = \{Short, Middle, Long\}$, representing different option time to maturity. $\Delta CallIV_t^D$ and $\Delta PutIV_t^D$ represent the changes in call and put option implied volatility within option category D on day t, respectively. $CallOIB_t^D$ and $PutOIB_t^D$ denote the call and put order imbalance within option category D, respectively. And $Return_t$ and $Amount_t$ represent the open-to-close return and trading amount of underlying ETF, respectively. All variables except $Amount_t$ are expressed in percentage terms, and $Amount_t$ is expressed in units of ten million. The Newey-West standard errors are reported in parentheses. *, **, *** represent 10%, 5%, and 1% significance levels, respectively. The sample period is from February 9, 2015, to February 9, 2025.

TABLE A4 | Recursively out-of-sample R^2 and trading strategy based on pure predictive regression.

	Out of sample R^2			Average return of trading strategy		
	Model 1	**Model 2**	**Model 3**	**Model 1**	**Model 2**	**Model 3**
call	−0.62			−0.03(0.06)		
put		−0.73			−0.01(0.05)	
call and put			−0.63			−0.01(0.05)

Note: The table reports the recursively out-of-sample R^2 and trade strategy returns based on pure predictive regression. The out-of-sample forecasting procedure consists of two steps. Step 1, estimating the predictive regression model:

$$\begin{aligned}\text{Return}_t = \alpha + \beta_1\text{CallOIB}_{t-1} + \gamma_1\text{PutOIB}_{t-1} + \lambda\text{Controls}_{t-1}\\ + \phi\text{Return}_{t-1} + \varepsilon_t.\end{aligned}$$

using data from the training period, 1, ..., $t-1$. Step 2, Obtaining the predicted return for period t by substituting the observable variables at period $t-1$ into the model. The control variables include TermSpread, MarketHV, Turnover, ILLIQ, PCR, OSR, IVS, SKEW, POV, and NOV. The out-of-sample R^2 is computed as

$$R^2_{\text{OS}} = 1 - \frac{\sum(\text{Return}_s - \hat{r}_s)^2}{\sum(\text{Return}_s - \bar{r}_s)^2},$$

where Return_s is the actual daily open-to-close return of underlying in testing period s, and $\hat{r}_s$ and $\bar{r}_s$ are the out-of-sample forecasted return and the historical average return, respectively. The trading strategy is that if the forecasted return $\hat{r}_s > 0$, buy one share of the underlying at the open price and sell it at the close price. The strategy return is defined as,

$$\text{Return}_s^{\text{trading}} = \begin{cases}\text{Return}_s, & \text{if} \quad \hat{r}_s > 0\\ 0, & \text{if} \quad \hat{r}_s \leq 0\end{cases}.$$

Model 1 and Model 2 represent that in step 1 the regression model contains call and put order imbalance separately, and Model 3 represent that the model contains call and put order imbalance simultaneously. The full sample period is from February 9, 2015, to February 9, 2025, and is divided into two segments. The period from February 9, 2015, to February 8, 2022, serves as the initial training period. The period from February 9, 2022, to February 9, 2025, constitutes the out-of-sample testing period.